

Soft Voting for Policy-Aware Private Data Synthesis

Yingge Hu

yhu893@uwo.ca

Department of Computer Science,
Western University
London, Ontario, Canada

Gautham Ramesh Babu

grg2@uwo.ca

Department of Computer Science,
Western University
London, Ontario, Canada

Mostafa Milani

mostafa.milani@uwo.ca

Department of Computer Science,
Western University
London, Ontario, Canada

Abstract

Blowfish privacy relaxes differential privacy (DP) by protecting only the attribute-value substitutions a data owner specifies as edges of a policy graph. A sparser policy can reduce the noise required by a mechanism, but only when the released statistic changes less across protected substitutions than across arbitrary DP neighbors. We study this question for evolutionary, nearest-neighbor DP synthesizers such as Private Evolution (PE) and its tabular instantiation Tab-PE, which score private records against a candidate population and release a noisy vote histogram. Their *hard* vote is constant inside each candidate’s decision region and jumps at its boundary. Its policy-specific sensitivity therefore equals the full worst-case value whenever at least one protected substitution crosses a boundary, regardless of how short that substitution is. Because every round we examined contained such a substitution, the policy graph gave no reduction in noise.

We propose BF-Soft, a temperature-smoothed *soft* vote whose response changes gradually with distance. Its sensitivity has a tight closed-form bound in the policy graph’s reach and the temperature, independent of the number of candidates, and the bound can be computed once before synthesis. It also predicts from the policy alone when policy-aware smoothing cannot substantially reduce noise: protecting a flat categorical or binary attribute drives the reach to its maximum. On real and synthetic datasets under narrow numeric policies, BF-Soft reduces error relative to hard voting at strong privacy budgets, while the advantage reverses at weaker budgets. A public-data pilot predicts when soft voting is beneficial without spending private budget.

Keywords

Differential privacy, Blowfish privacy, Policy-aware privacy, Synthetic data generation, Private Evolution, Sensitivity analysis

ACM Reference Format:

Yingge Hu, Gautham Ramesh Babu, and Mostafa Milani. 2027. Soft Voting for Policy-Aware Private Data Synthesis. In *Proceedings of 30th International Conference on Extending Database Technology (EDBT 2027)*. ACM, New York, NY, USA, 22 pages. <https://doi.org/10.1145/nnnnnnnn.nnnnnnnn>

1 Introduction

Differentially private (DP) synthetic data lets a data owner share a sensitive table while limiting how much the released output can change when one individual’s record changes [6]. Under bounded substitution adjacency, every allowed replacement of one record is protected by the same worst-case privacy guarantee. On large or structured domains such as age, income, or ZIP codes, this worst-case symmetry can contribute substantially to the utility gap between private and non-private synthetic data.

Policy-based definitions relax this symmetry by allowing the data owner to specify which distinctions require protection. Pufferfish [14] makes the protected secrets explicit, and Blowfish privacy [10] instantiates this idea for tabular data using a *policy graph*: an edge between two values marks a substitution that the mechanism must protect. A policy might, for example, protect a change from age 34 to 35 while not requiring the same protection between ages 34 and 70. A policy graph therefore restricts the neighbor relation used to compute sensitivity, so a release’s policy-specific sensitivity never exceeds its standard global sensitivity [10]. The inequality can, however, be tight. A sparser graph improves utility only when the released statistic changes less across protected edges than across arbitrary DP neighbors. This occurs for structured linear workloads such as range and marginal queries [9], but not for a function whose response remains at its worst case even on short protected substitutions.

We study when this distinction matters for evolutionary, nearest-neighbor DP synthesis. Rather than privately measuring a fixed set of statistics, this paradigm reshapes a population of candidate synthetic records round by round. Each real record *votes* for a candidate; the votes are aggregated into a histogram, noise is added, and the most-supported candidates survive while the rest are replaced. This voting step, introduced in Private Evolution [16] (PE) and extended to tabular data by Tab-PE [24], is where privacy noise enters. Recent extensions modify the vote or the protection target along other axes (Section 3), but they do not ask whether the vote’s response can reflect how close a protected substitution is under a policy graph.

This question is important because a sparse policy graph is useful only if the mechanism can exploit it. We therefore ask: *when does a Blowfish policy reduce the sensitivity of the nearest-neighbor vote, and can the vote be redesigned so that its sensitivity changes gradually with the policy?*

The answer depends on the vote itself, not on the mutation rule, the number of rounds, or the domain. The standard *hard* vote assigns a record to its single nearest candidate. A protected substitution therefore has only two effects: either the nearest candidate remains the same, and the vote does not change, or the assignment crosses a decision boundary, and the vote moves entirely to another candidate. In the latter case the ℓ_2 change is the full worst-case value $\sqrt{2}$, regardless of how short the protected substitution is. Thus the hard vote cannot exploit the *degree* to which a policy restricts substitutions: its policy-specific sensitivity is either 0 or $\sqrt{2}$. In every round we checked in our experiments, some protected edge crossed a decision boundary, so the policy graph reduced no noise.

We replace the hard vote with BF-Soft, a temperature-smoothed *soft* vote that splits each record’s vote across candidates according to a softmax of negative distance. A short substitution then shifts only

a small part of the vote, while larger substitutions can shift more. We show that its policy-specific sensitivity is at most $\sqrt{2} \tanh(r/2\tau)$, where r is the policy graph’s reach, the largest distance between two protected values, and τ is the temperature. The bound grows with the reach but never exceeds the hard-vote value $\sqrt{2}$, so a policy protecting only short substitutions yields less noise. It is tight, independent of the number of candidates, and computed once before synthesis. Our proof builds on a total-variation inequality for exponential tilts [5]: softmax ignores a common shift in its scores, so only their relative change matters, and the reach bounds it for all candidates at once.

The result also clarifies when policy-aware smoothing cannot help. Reach is a maximum over protected attributes rather than a sum, so adding another narrowly protected numeric attribute need not increase the bound. In contrast, protecting a flat categorical or binary attribute contributes that attribute’s maximum distance and can make the bound essentially equal to the hard vote’s worst case at useful temperatures. Participation (add/remove) edges similarly impose a floor that substitution policies cannot remove. Temperature creates a second trade-off: increasing τ reduces sensitivity but also flattens the vote and weakens the evolutionary selection signal. The bounds therefore do not determine an optimal temperature, and we use a public-data pilot to select it without spending private budget.

Empirically, BF-Soft reduces error relative to hard voting at strong privacy budgets under narrow numeric policies. On the UCI Adult and Bank benchmarks, the crossover occurs at approximately $\epsilon = 0.2$ – 0.45 at $\delta = 10^{-5}$ for the fixed temperatures we study; beyond this region, the cost of smoothing can dominate the saved noise. Our experiments also isolate these two effects, test policies protecting multiple attributes, and compare the analytical sensitivity bound with exact sensitivity.

Our contributions are as follows.

- *A diagnosis of hard voting under policies.* For a linear query under unconstrained bounded Blowfish adjacency, policy-specific sensitivity reduces to a maximum over the policy graph’s edges. For the hard vote this maximum is all-or-nothing: it jumps from 0 to $\sqrt{2}$ at the first protected boundary crossing (Section 5).
- *BF-Soft, a soft vote whose policy-specific sensitivity has a tight closed-form bound in the policy’s reach and the temperature.* The bound is independent of the candidate count, can be computed once before synthesis, and identifies policies for which smoothing cannot substantially reduce noise (Sections 4 and 5).
- *An analysis of the temperature trade-off.* We bound both its effect on sensitivity and its effect on the vote’s ability to discriminate among candidates, and use a public-data pilot to select the temperature without spending private budget.
- *An evaluation on the UCI Adult and Bank benchmarks and XOR, a synthetic generator with planted higher-order (XOR) dependencies, that locates where policy-aware soft voting helps and where the advantage reverses, across privacy budgets and single- and multi-attribute policies.* The experiments separate the benefit of lower noise from the cost of changing the vote and evaluate the tightness and cost of the calibration bound (Section 6).

Sections 2 and 3 review background and related work, respectively, Section 4 presents the construction, Section 5 its analysis, and Section 6 the evaluation. The extended version with all appendices is available in [11], and our code in [12].

2 Preliminaries

We review the privacy definitions and synthesis paradigm we use.

2.1 Differential Privacy and zCDP

A database D is a multiset of n records from a discrete domain $\mathcal{T}$, with histogram $h(D) \in \mathbb{N}^{|\mathcal{T}|}$. Databases D, D' are *neighbors*, written $D \sim D'$, if they differ in one record. Under *bounded* adjacency, n is public and one record’s value changes. Under *unbounded* adjacency, one record is added or removed.

Definition 2.1 (Differential Privacy [6]). A randomized mechanism $\mathcal{M}$ satisfies ϵ -DP if for all neighboring $D \sim D'$ and all measurable $S \subseteq \text{Range}(\mathcal{M})$, $\Pr[\mathcal{M}(D) \in S] \leq e^\epsilon \Pr[\mathcal{M}(D') \in S]$.

For $f : \mathcal{T}^n \rightarrow \mathbb{R}^d$, the *global sensitivity* $\Delta_p f = \max_{D \sim D'} \|f(D) - f(D')\|_p$ calibrates the noise added to $f(D)$. Laplace noise scaled to $\Delta_1 f / \epsilon$ gives ϵ -DP, and Gaussian noise with $\sigma^2 = (\Delta_2 f)^2 / (2\rho)$ gives ρ -zCDP. We account for privacy with zero-concentrated DP [3].

Definition 2.2 (zCDP [3]). A mechanism $\mathcal{M}$ satisfies ρ -zCDP if for all neighboring $D \sim D'$ and all $\alpha \in (1, \infty)$, the Rényi divergence of order α between $\mathcal{M}(D)$ and $\mathcal{M}(D')$ is at most $\rho\alpha$.

zCDP is closed under post-processing and composes additively: an adaptively chosen sequence of mechanisms with budgets $\rho_1, \dots, \rho_k$ satisfies $\sum_i \rho_i$ -zCDP. A ρ -zCDP mechanism satisfies (ϵ, δ) -DP for every $\delta > 0$ with [3]

$$\epsilon = \rho + 2\sqrt{\rho \ln(1/\delta)}. \quad (1)$$

The synthesizer we study computes each round’s candidates from earlier noisy outputs. We therefore analyze each round conditional on its candidate population and compose adaptively (Section 5.1).

2.2 Pufferfish and Blowfish Privacy

Pufferfish [14] replaces DP’s fixed neighbor relation with an explicit set of protected secret pairs. Blowfish [10] instantiates this idea for tabular data with a policy graph.

Definition 2.3 (Policy graph [10]). A policy is a triple $P = (\mathcal{T}, G, I_Q)$, where $G = (V, E)$ with $V \subseteq \mathcal{T} \cup \{\perp\}$ is the *policy graph*. An edge $(u, v) \in E$ marks values u, v that must remain indistinguishable. An edge $(u, \perp)$ to a dummy vertex marks the presence of a record with value u as protected. I_Q is the set of databases consistent with public deterministic constraints Q .

Throughout, I_Q is unconstrained, and every result in Section 5 assumes this.

Definition 2.4 (Policy neighbors and policy-specific sensitivity [10]). $D, D' \in I_Q$ are *policy neighbors*, $D \sim_P D'$, if they are minimally different and their difference corresponds to an edge of G . An edge (u, v) replaces one record’s value u by v , and an edge $(u, \perp)$ adds or removes a record with value u . Let $N(P)$ be the set of all such pairs. For $f : I_Q \rightarrow \mathbb{R}^d$ and $p \geq 1$, the policy-specific sensitivity is

$$S_p(f, P) = \max_{(D, D') \in N(P)} \|f(D) - f(D')\|_p.$$

Definition 2.5 (Blowfish privacy [10] and its zCDP form). A mechanism satisfies (ϵ, P) -Blowfish privacy if the condition of Definition 2.1 holds for all $D \sim_P D'$. It satisfies (ρ, P) -zCDP Blowfish privacy if the condition of Definition 2.2 holds for all $D \sim_P D'$.

The proofs of post-processing, composition, and Eq. (1) use only the divergence bound for a fixed neighboring pair, never the structure of the neighbor relation. All three therefore carry over unchanged to (ρ, P) -zCDP Blowfish privacy. In particular, T rounds with budgets ρ_t give $(\sum_t \rho_t, P)$ -zCDP Blowfish privacy, which implies (ϵ, δ, P) -Blowfish privacy with ϵ as in Eq. (1).

THEOREM 2.6 ([3, 10]). *Releasing $f(D) + \eta$ with $\eta_i \sim \text{Lap}(S_1(f, P)/\epsilon)$ i.i.d. satisfies (ϵ, P) -Blowfish privacy. Releasing $f(D) + \mathcal{N}(0, \sigma^2 I)$ with $\sigma^2 = S_2(f, P)^2/(2\rho)$ satisfies (ρ, P) -zCDP Blowfish privacy. When G is the complete graph on $\mathcal{T}$ and I_Q is unconstrained, both coincide with their standard bounded-DP counterparts.*

Since $N(P)$ contains only standard neighbor pairs, $S_p(f, P) \leq \Delta_p f$ for every G . For the sensitivity-calibrated mechanisms we use, a Blowfish policy reduces noise only when this inequality is strict. A sparse policy graph is therefore not sufficient by itself: the released function must also respond less to the substitutions represented by its edges than to arbitrary DP neighbors. Section 5 characterizes this condition for the voting functions we study.

Values not joined by an edge are still protected along paths of G . A shortest path of length ℓ gives $\ell\epsilon$ under (ϵ, P) -Blowfish privacy and, by group privacy for zCDP [3], $\ell^2\rho$ under (ρ, P) -zCDP Blowfish privacy. Values in different connected components are not protected from each other [10].

Specifying policies. A data owner typically specifies one graph per attribute of a product domain $\mathcal{T} = \mathcal{T}_1 \times \dots \times \mathcal{T}_p$. These combine into a record-level graph G that links two records agreeing on every attribute but one, on which they differ by a protected pair. Our experiments (Section 6) use three common forms:

- *Threshold graphs* protect nearby values under a numeric, ordinal, or hierarchical distance: values within distance θ are adjacent, while more distant values are not directly protected (Definition 2.7; the hierarchical case is in Appendix D).
- *Block-partition graphs* protect substitutions within a coarse category but not across categories. They can, for example, hide which value inside a category an individual has while allowing the category itself to remain distinguishable.
- *Complete graphs* connect every pair of values of an attribute and therefore protect that attribute without distinguishing among its values.

Edges to $\perp$ express unbounded adjacency in the same formalism; which values receive them is a modeling choice (Section 5.6). Our experiments use bounded adjacency. Results in Section 5 hold for any G unless stated otherwise.

Definition 2.7 (Threshold policy graph and graph reach). Let d be a distance on $\mathcal{T}$. For $\theta \geq 0$, the *threshold policy graph* $G_\theta = (\mathcal{T}, E_\theta)$ has edge set $E_\theta = \{(u, v) : 0 < d(u, v) \leq \theta\}$. For any policy graph $G = (\mathcal{T}, E)$, the *graph reach* with respect to d is $r_d(G) = \max_{(u, v) \in E} d(u, v)$, so $r_d(G_\theta) = \theta$ whenever some pair attains the threshold.

Graph reach summarizes the longest substitution that the policy protects under the same distance used by the vote. It does not describe the entire topology of G , but Section 5.4 shows that it is sufficient for a population-independent sensitivity bound for our soft vote.

2.3 Evolutionary, Nearest-Neighbor Synthesis

Private Evolution [16] (PE) and its tabular instantiation Tab-PE [24] evolve a population of candidate records rather than measuring marginals of the data. The initial population is generated without private data. At round t , with population $C_t = \{c_1, \dots, c_m\}$, each private record $x \in D$ votes for its nearest candidate under a base distance d (e.g., Gower distance for mixed attributes):

$$v_{C_t}(x) = e_{\text{nn}(x)} \in \{0, 1\}^m, \quad \text{nn}(x) = \arg \min_j d(x, c_j),$$

where e_j is the j th standard basis vector. This is the DP_NN_HISTOGRAM step of PE and Tab-PE in vector form. The released statistic is the histogram $H_t(D; C_t) = \sum_{x \in D} v_{C_t}(x)$, noised as $\tilde{H}_t = H_t(D; C_t) + \mathcal{N}(0, \sigma_t^2 I)$ under a per-round budget ρ_t . A resampling and mutation step, `ResampleMutate`, reads only $\tilde{H}_t$ and C_t to form C_{t+1} , and the loop repeats for T rounds.

Our analysis (Sections 4 and 5) depends only on the form of v_{C_t} , not on d , the mutation rule, or T . We evaluate it in a Tab-PE-style instantiation (Section 6). Because v_{C_t} is one-hot, its response to a substitution takes only two values. Section 5.3 shows that this prevents sensitivity from varying gradually with the policy's notion of closeness.

3 Related Work

We position our work against DP synthesizers, policy-aware mechanisms, metric relaxations of DP, and smoothed assignment rules.

Evolutionary synthesis PE [16] evolves a candidate population through private nearest-neighbor voting against the real data, without training a model. Its tabular instantiation Tab-PE [24] reports gains over marginal-based baselines on data with high-order correlations that low-order marginals miss. Both use a one-hot vote each round. Under a policy graph, this vote remains constant while the nearest candidate is unchanged and jumps when a substitution crosses a decision boundary, which makes its policy-specific sensitivity a step function (Section 5.3).

Two extensions relax PE along axes orthogonal to ours. WASP [30] spreads a record's vote over its top- Q candidates with weights determined by rank. This makes the vote less concentrated, but rank is itself piecewise constant in the underlying distances: a protected substitution can leave every rank unchanged or change a rank discontinuously at a boundary. Its response is therefore still step-like rather than graded by the substitution distance. In contrast, our soft vote changes continuously with distance. WASP changes how many candidates share a vote; we change how the vote responds to a substitution, and the two could in principle compose. Appendix A gives the fuller comparison.

SecPE [25] relaxes PE's protection target rather than its vote, replacing membership privacy with a secret-specific (p, r) -protection guarantee; it has no policy graph.

Other DP synthesizers Marginal-based mechanisms such as PrivBayes [29], Private-PGM [20], MST [18], and AIM [19] privately measure selected low-order marginals and fit a joint distribution to them. GAN-based mechanisms [13, 27, 28] add noise during training rather than to a fixed statistic. Both families target standard DP. Our goal is to isolate how a released statistic can exploit a policy within one synthesis paradigm. A utility comparison across paradigms under a common policy would require policy-aware versions of the competing synthesizers, which we leave open.

Policy-aware mechanisms Haney et al. [9] answer a fixed linear workload under Blowfish by transforming the workload and policy graph so that standard DP mechanisms apply. Conditional on C_t , one round’s vote histogram is itself a linear workload, and our sensitivity identity (Lemma 5.1) is consistent with this view. The difference is where the design freedom lies. Their workload is given and the mechanism is optimized for answering it. In evolutionary synthesis, the per-record voting map determines the workload and the candidate population changes adaptively across rounds. We redesign this map itself so that its response to a protected substitution reflects the policy’s distance, rather than only recalibrating noise for a fixed hard-vote workload.

Xiao et al. [26] study the complementary problem of when temporal correlation undermines the protection intended by a policy graph.

Metric relaxations Blowfish is one instance of relaxing DP’s neighbor relation according to an application-specific notion of closeness, as in metric DP [4], Lipschitz privacy [15], and geo-indistinguishability [1]. Geo-indistinguishability, for example, uses a mechanism whose output distribution varies continuously with geographic distance. Ball-DP [17] restricts substitutions to a fixed radius, which corresponds to a threshold policy graph (Definition 2.7), but applies the restriction to convex empirical risk minimization rather than to voting or synthesis. The fuller distinction is given in Appendix A.

Smoothed assignment rules Temperature-scaled softmax is a standard smooth relaxation of a hard arg max or arg min rule, and related sensitivity or Lipschitz bounds appear in the exponential mechanism [7, 21] and in bounds on the softmax map itself [23]. Applied directly to our logits, a generic ℓ_2 argument controls the output change by the ℓ_2 size of the logit perturbation. Although each coordinate changes by at most $d(x, y)/\tau$, this norm can be as large as $\sqrt{m} d(x, y)/\tau$ for m candidates.

Our analysis exploits additional structure of a policy substitution. Softmax is invariant to a common shift of all logits, so only their *oscillation* matters. A substitution from x to y has oscillation at most $2d(x, y)/\tau$, independent of m . Combining this observation with the total-variation inequality of Cohen and Fausti [5] yields the tight, saturating bound of Corollary 5.5; Proposition 5.6 shows that no uniformly smaller bound depending only on policy reach and temperature is possible. Appendix A gives the detailed comparison with generic softmax bounds.

4 Policy-Aware Evolutionary Synthesis

We replace the hard nearest-neighbor vote with a policy-aware alternative while leaving the rest of the evolutionary synthesis paradigm unchanged.

4.1 Generalizing the Vote

The hard vote $v_{C_t}(x) = e_{\text{nn}(x)}$ puts all of a record’s mass on its nearest candidate, a vertex of the probability simplex Δ^{m-1} over the m candidates in C_t . Neither noise calibration nor resampling requires the vote to be one-hot: both only need the per-record responses to aggregate into

$$H_t(D; C_t) = \sum_{x \in D} v_{C_t}(x).$$

We therefore allow any simplex-valued response,

$$v_{C_t}(x) \in \Delta^{m-1} = \{p \in \mathbb{R}_{\geq 0}^m : \sum_j p_j = 1\},$$

of which the hard vote is the special case confined to the simplex vertices.

This abstraction separates the *per-record response* from the rest of the evolutionary loop. Once the responses are aggregated and privately released, candidate resampling and mutation are post-processing. We can therefore change the vote and its sensitivity calibration without changing the remainder of the synthesis procedure.

4.2 Blowfish-Compatible Voting

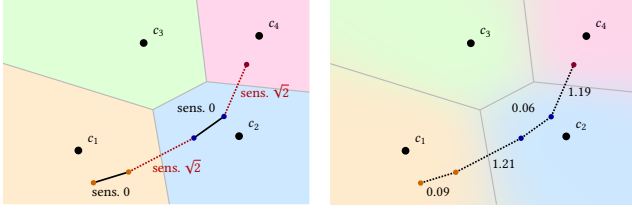
We define BF-Evo as the evolutionary paradigm run under a policy P . It replaces standard adjacency with $\sim_P$ and calibrates each round’s noise to the policy-specific sensitivity of that round’s histogram, for any simplex-valued vote. At round t , conditional on C_t , it releases

$$\tilde{H}_t = H_t(D; C_t) + \mathcal{N}(0, \sigma_t^2 I), \quad \sigma_t^2 = \frac{S_2(H_t, P)^2}{2\rho_t},$$

where $S_2(H_t, P)$ is taken over the edges of G with C_t fixed. By Theorem 2.6 and adaptive composition (Section 2), T rounds satisfy $(\sum_t \rho_t, P)$ -zCDP Blowfish privacy, and hence (ϵ, δ, P) -Blowfish privacy via Eq. (1).

This gives two distinct ways to make the evolutionary mechanism policy-aware. We can keep the original hard vote and only replace its global sensitivity by its policy-specific sensitivity; we call this method *Hard*. Or we can also redesign the per-record response so that protected substitutions induce smaller changes in the vote; this gives BF-Soft below. Tab-PE, Hard, and BF-Soft therefore share the same evolutionary loop but differ in their vote and/or its calibration.

We use bounded adjacency throughout. Participation edges, which encode add/remove adjacency in the same graph, are treated in Section 5.6. Practical PE implementations also zero out small entries of $\tilde{H}_t$ with a public threshold. This is post-processing and does not affect sensitivity (Appendix B).



(a) Hard vote: 0 inside a cell, $\sqrt{2}$ across a boundary. (b) Soft vote ($\tau = 0.5$): grades smoothly to the $\sqrt{2}$ ceiling.

Figure 1: Sensitivity along the same threshold policy graph and candidate population, hard vote (left) vs. soft vote (right).

4.3 Hard and Soft Votes

Hard vote A protected substitution either keeps a record’s vote on the same candidate, at no cost, or moves it entirely to another candidate. In the second case it incurs the full ℓ_2 distance $\sqrt{2}$ between two one-hot vectors, regardless of how short the substitution is.

Figure 1a shows this on C_t ’s Voronoi cells: solid policy edges whose endpoints lie in the same cell cost 0, while the dotted edge crossing a cell boundary costs $\sqrt{2}$. Thus geometric closeness of a protected pair does not translate into closeness of the released response once the pair crosses a decision boundary. Corollary 5.3 makes this exact: the hard vote’s sensitivity is $\sqrt{2}$ whenever any protected edge crosses a boundary, and 0 otherwise.

Whether a crossing exists can change from round to round as the candidate population evolves (Section 5.3); Appendix C gives the analogue for other adjacency notions. In every round we checked in our experiments, some protected edge crossed a boundary, so Hard’s calibration equaled the raw $\sqrt{2}$ of Tab-PE (Section 6).

Soft vote BF-Soft replaces the one-hot assignment with a temperature-smoothed vote under the same base distance d :

$$v_{C_t}(x)[j] = \frac{\exp(-d(x, c_j)/\tau)}{\sum_k \exp(-d(x, c_k)/\tau)}. \quad (2)$$

For $\tau > 0$, the response varies continuously with the candidate distances. A short substitution can therefore move only a small amount of voting mass, while a larger substitution can move more, up to the $\sqrt{2}$ diameter attained by two different one-hot votes. Figure 1b illustrates this graded response on the same policy graph used for the hard vote.

Sensitivity is still the worst response over all protected pairs, but the response to an individual edge is no longer restricted to $\{0, \sqrt{2}\}$. Section 5.4 shows that, for every candidate population,

$$S_2(H_t, P) \leq \sqrt{2} \tanh\left(\frac{r_d(G)}{2\tau}\right).$$

The bound decreases smoothly as the policy reach $r_d(G)$ shrinks or the temperature τ grows, and is independent of the number of candidates. Unlike exact sensitivity, it also does not depend on the realized population C_t .

Categorical attributes The bound measures policy edges using the same base distance d that defines the vote. For numeric and ordinal attributes, a policy can therefore restrict the permitted substitution distance directly. Under Gower’s default categorical term, however,

Table 1: The three instantiations of Algorithm 1.

Method	Per-record vote	Calibration	Recomputed each round?
Tab-PE	one-hot	$\sqrt{2}$	no
Hard/BF-Evo	one-hot	0 or $\sqrt{2}$	yes, for exact calibration
BF-Soft	softmax	$\sqrt{2} \tanh(r_d(G)/2\tau)$	no

every pair of distinct values has the same nonzero distance. Protecting a flat categorical or binary attribute consequently contributes its full Gower weight to the reach, the largest contribution that attribute can make.

The analytical bound remains strictly below $\sqrt{2}$ for finite reach and $\tau > 0$ (Remark 1), but at temperatures that retain a sharp selection signal it can be numerically very close to $\sqrt{2}$, leaving little noise reduction. When a categorical attribute has genuine hierarchical structure, replacing the flat distance by a tree metric restores several levels of policy distance; we discuss this case in Appendix D.

Temperature The temperature τ affects more than sensitivity. Increasing it flattens the vote toward the uniform distribution and weakens the selection pressure available to ResampleMutate. It therefore trades privacy-noise reduction against the usefulness of the voting signal. Section 5.5 makes both limits explicit, and Section 6 measures the resulting utility trade-off.

4.4 Instantiation and Calibration Cost

BF-Evo and BF-Soft change only the per-record vote and its calibration, so they can be substituted into any implementation of the evolutionary paradigm. In our Tab-PE-style instantiation, d is a Gower distance and the mutation operator is tuned for tabular data. The resampling rule, mutation schedule, candidate populations, and privacy-budget schedule are otherwise unchanged.

Algorithm 1 makes the comparison explicit. The three methods share the aggregation, Gaussian mechanism, and ResampleMutate steps. They differ only at the vote (line 3) and the sensitivity calibration (line 11).

Table 1 summarizes the resulting differences. Tab-PE and Hard use exactly the same one-hot response; Hard differs only by restricting the sensitivity maximum to policy edges. BF-Soft changes the response itself and calibrates to a population-independent upper bound.

Round-dependent versus closed-form calibration For Hard, exact calibration depends on the current population C_t . One protected edge whose endpoints have different nearest candidates is enough to certify sensitivity $\sqrt{2}$, but certifying sensitivity 0 requires ruling out such a crossing for every protected edge. Since resampling and mutation change C_t , the same fixed policy can give different exact sensitivities in different rounds.

For BF-Soft, the exact sensitivity also depends on C_t , and can be substantially smaller than the analytical bound. The difference is that the bound depends only on the policy reach $r_d(G)$ and the fixed temperature τ . It can therefore be computed once before synthesis and remains valid for every candidate population encountered by the adaptive process.

Algorithm 1 Evolutionary DP Synthesis

Input: policy graph $G = (\mathcal{T}, E)$, reach $r_d(G)$, distance d , temperature τ , budgets $\{\rho_t\}_{t=1}^T$, method $\in \{\text{Tab-PE}, \text{BF-Evo}, \text{BF-Soft}\}$
Output: final candidate population C_{T+1}

```

1:  $C_1 \leftarrow \text{InitPopulation}()$ ;
2: for  $t = 1, \dots, T$  do
3:   for  $x \in D$  do
4:     if method = BF-Soft then
5:        $v_{C_t}(x) \leftarrow \text{softmax}_j(-d(x, c_j)/\tau)$ ;  $\triangleright$  Eq. (2)
6:     else
7:        $v_{C_t}(x) \leftarrow e_{\arg \min_j d(x, c_j)}$ ;  $\triangleright$  nn(x)
8:     end if
9:   end for
10:   $H_t(D; C_t) \leftarrow \sum_{x \in D} v_{C_t}(x)$ ;
11:  if method = Tab-PE then
12:     $\Delta_t \leftarrow \sqrt{2}$ ;  $\triangleright$  global sensitivity
13:  else if method = BF-Evo then
14:     $\Delta_t \leftarrow \sqrt{2} \cdot \mathbb{1}[\exists (u, v) \in E : \text{nn}_{C_t}(u) \neq \text{nn}_{C_t}(v)]$ ;  $\triangleright$ 
    Cor. 5.3
15:  else if method = BF-Soft then
16:     $\Delta_t \leftarrow \sqrt{2} \tanh(r_d(G)/2\tau)$ ;  $\triangleright$  Cor. 5.5
17:  end if
18:   $\sigma_t^2 \leftarrow \Delta_t^2/(2\rho_t)$ ;
19:   $\tilde{H}_t \leftarrow H_t(D; C_t) + N(0, \sigma_t^2 I)$ ;
20:   $C_{t+1} \leftarrow \text{ResampleMutate}(\tilde{H}_t, C_t)$ ;
21: end for
22: return  $C_{T+1}$ ;

```

Why not calibrate to exact soft sensitivity? Because C_t is itself obtained from previous private releases, conditioning on the realized C_t and computing exact sensitivity would not by itself violate privacy. The obstacle is computational. By Lemma 5.1, exact sensitivity requires maximizing the change in the per-record response over every edge of the *record-level* policy graph. For a product domain, a protected pair on one attribute appears once for every combination of the remaining attributes. Thus the record-level edge set can grow exponentially in the number of attributes even when each per-attribute policy is simple.

The same search is needed to certify Hard’s zero-sensitivity case. By contrast, once $r_d(G)$ has been obtained from the per-attribute policies, evaluating the BF-Soft bound is constant-time and requires no inspection of C_t . Both hard nearest-neighbor voting and softmax voting cost $O(m)$ per record, so the soft vote changes no asymptotic voting cost. Appendix C gives the full adjacency and calibration-cost discussion, and Section 6.2.6 evaluates the cost empirically.

5 Sensitivity of Hard and Soft Votes

We now make Section 4’s claims precise. For any linear query, policy-specific sensitivity reduces to a maximum over the policy graph’s edges (§5.2). For the hard vote this maximum is all-or-nothing (§5.3), while for the soft vote it admits a tight closed-form bound in the policy reach and temperature (§5.4). Temperature trades this reduction against the vote’s ability to discriminate among

candidates (§5.5), and some policy edges impose floors that reach-dependent calibration cannot lower (§5.6). Full proofs are in Appendix E.

5.1 Linear Queries

Let $(\mathcal{T}, d)$ be a finite metric space and $\phi : \mathcal{T} \rightarrow \mathbb{R}^m$ a per-record embedding. A *linear query* [9] is a release

$$f(D) = \sum_{x \in D} \phi(x) = W h(D),$$

linear in the database histogram $h(D)$ even when ϕ is nonlinear in x .

At a fixed round, conditional on C_t and τ , both votes of Section 4 are linear queries. The hard vote has $\phi(x) = e_{\text{nn}(x)}$, while the soft vote has $\phi(x) = v_{C_t}(x)$ from Eq. (2). Raw marginal counts are linear queries as well. Model-fitting procedures optimized jointly over all records, such as Private-PGM [20], are not of this form.

All results below are therefore per-round statements conditional on the realized C_t . This is sufficient for privacy accounting because C_t is computed from earlier private releases, and the rounds compose adaptively. The policy P itself is fixed before the mechanism runs.

5.2 A Sensitivity Identity

Replacing one record’s value u by v changes a linear query by exactly $\phi(u) - \phi(v)$; every other record cancels. Thus, under bounded and unconstrained Blowfish adjacency, the worst case over databases becomes a worst case over policy edges.

LEMMA 5.1 (POLICY SENSITIVITY IS A MAXIMUM OVER EDGES). *For a linear query $f(D) = \sum_{x \in D} \phi(x)$, any policy graph $G = (\mathcal{T}, E)$, unconstrained I_Q , and bounded adjacency,*

$$S_p(f, P) = \max_{(u, v) \in E} \|\phi(u) - \phi(v)\|_p,$$

with value 0 when $E = \emptyset$.

PROOF SKETCH. A policy neighbor replaces one value u by v with $(u, v) \in E$, so linearity gives $f(D) - f(D') = \phi(u) - \phi(v)$. Conversely, every policy edge can be realized by such a neighboring pair. Taking the maximum gives the claim. $\square$

This identity is elementary and consistent with the workload view of Haney et al. [9]; its role here is to isolate the record-level response that the evolutionary mechanism itself is free to redesign.

COROLLARY 5.2 (ATTRIBUTE-BY-ATTRIBUTE DECOMPOSITION). *For a product domain $\mathcal{T}_1 \times \dots \times \mathcal{T}_p$ with per-attribute graphs $G_i = (\mathcal{T}_i, E_i)$ and the record-level product graph G of Section 2,*

$$S_p(f, P) = \max_i S_{p,i}(f, P),$$

where $S_{p,i}$ ranges only over edges that change attribute i . Under a Gower-style distance $d(x, y) = \sum_i w_i d_i(x_i, y_i)$,

$$r_d(G) = \max_i w_i r_{d_i}(G_i).$$

PROOF SKETCH. Every record-level edge changes exactly one attribute. The edge set is therefore a disjoint union of the per-attribute edge sets, and an edge changing attribute i has distance $w_i d_i$. $\square$

A data owner can consequently reason about each attribute’s policy in isolation. Protecting additional attributes increases the analytical reach only when one of them has a larger weighted reach than those already protected.

5.3 Hard-Vote Sensitivity: A Step Function

COROLLARY 5.3 (HARD-VOTE SENSITIVITY IS SET BY BOUNDARY CROSSINGS). *For the hard vote $\phi(u) = e_{nn(u)}$ and any policy graph $G = (\mathcal{T}, E)$,*

$$S_2(f, P) = \begin{cases} \sqrt{2}, & \exists (u, v) \in E : nn(u) \neq nn(v), \\ 0, & \text{otherwise.} \end{cases}$$

PROOF SKETCH. If the nearest candidates agree, the two one-hot vectors are identical. Otherwise their difference has one +1 and one −1 entry and hence ℓ_2 norm $\sqrt{2}$. Lemma 5.1 takes the maximum over policy edges. $\square$

A graph whose edges all remain inside the same Voronoi cells therefore gives sensitivity 0. What the hard vote lacks is any graded dependence on policy distance: its response jumps directly from 0 to $\sqrt{2}$ at the first protected boundary crossing. In particular, once an edge crosses a boundary, shortening that edge does not reduce sensitivity.

The crossing structure is also population-dependent. Define

$$r_{C_t} = \min_{nn(u) \neq nn(v)} d(u, v),$$

the shortest substitution that crosses a current decision boundary. As C_t changes through resampling and mutation, so can r_{C_t} . A fixed threshold policy can therefore yield sensitivity 0 in one round and $\sqrt{2}$ in another. Appendix C gives the corresponding values for add/remove and multi-row neighbors.

5.4 A Tight Soft-Vote Sensitivity Bound

The soft vote replaces this discontinuous response by a smooth one, but a generic softmax Lipschitz argument is not enough for our purpose. With logits

$$z(x)_j = -d(x, c_j)/\tau,$$

the triangle inequality gives, for every candidate j ,

$$|z(x)_j - z(y)_j| \leq \frac{d(x, y)}{\tau}.$$

A direct ℓ_2 bound on the logit perturbation can therefore scale as $\sqrt{m} d(x, y)/\tau$, introducing an undesirable dependence on the candidate count.

The key observation is that softmax is invariant to adding the same constant to every logit. Hence only the *relative* displacement of the logits matters. Writing $a = z(y) - z(x)$,

$$\text{osc}(a) := \max_j a_j - \min_j a_j \leq \frac{2d(x, y)}{\tau},$$

independent of m . The next lemma converts this oscillation bound into a sharp change in the resulting probability vector.

LEMMA 5.4 (TOTAL VARIATION UNDER BOUNDED LOG-TILTING; CF. [5]). *Let p be a distribution on $\{1, \dots, m\}$ and*

$$q_j = \frac{p_j e^{a_j}}{\sum_k p_k e^{a_k}}$$

with $\text{osc}(a) \leq s$. Then

$$\|p - q\|_1 \leq 2 \tanh(s/4),$$

and the bound is attained by a suitable two-point distribution.

PROOF IDEA. Because adding a constant to every a_j leaves q unchanged, center the perturbation so that $a_j \in [-s/2, s/2]$. Let $W_j = e^{a_j} \in [L, U]$ with $L = e^{-s/2}$ and $U = e^{s/2}$, and let $Z = \sum_j p_j W_j$. Then

$$\|p - q\|_1 = \frac{2}{Z} \mathbb{E}_p[(W - Z)^+].$$

Convexity of $w \mapsto (w - Z)^+$ bounds the expectation by the chord through L and U :

$$\mathbb{E}_p[(W - Z)^+] \leq \frac{(Z - L)(U - Z)}{U - L}.$$

Substitution gives

$$\|p - q\|_1 \leq \frac{2}{U - L} \left[(U + L) - \left(Z + \frac{1}{Z} \right) \right].$$

Since $LU = 1$, AM–GM shows that the right-hand side is maximized at $Z = 1$, where it equals $2 \tanh(s/4)$. The complete equality conditions and derivation are in Appendix E. $\square$

We now apply this result to the vote.

COROLLARY 5.5 (SENSITIVITY BOUND FOR BF-Soft). *For the soft-vote histogram $H_t(D; C_t)$ and any policy graph G ,*

$$S_2(H_t, P) \leq \sqrt{2} \tanh\left(\frac{r_d(G)}{2\tau}\right),$$

independent of $m = |C_t|$. For a threshold graph G_θ , this is

$$S_2(H_t, P_\theta) \leq \sqrt{2} \tanh(\theta/2\tau).$$

PROOF. By Lemma 5.1, it is enough to bound $\|v_{C_t}(x) - v_{C_t}(y)\|_2$ for one policy edge (x, y) . The vote at y is an exponential tilt of the vote at x , and the corresponding logit perturbation has oscillation at most $2d(x, y)/\tau$. Hence Lemma 5.4 gives

$$\|v_{C_t}(x) - v_{C_t}(y)\|_1 \leq 2 \tanh\left(\frac{d(x, y)}{2\tau}\right).$$

The difference of two probability vectors sums to zero, and any zero-sum vector z satisfies $\|z\|_2 \leq \|z\|_1/\sqrt{2}$. Therefore

$$\|v_{C_t}(x) - v_{C_t}(y)\|_2 \leq \sqrt{2} \tanh\left(\frac{d(x, y)}{2\tau}\right).$$

Maximizing over policy edges and using $d(x, y) \leq r_d(G)$ proves the claim. $\square$

The candidate count disappears because the proof controls the *oscillation* of the logit perturbation rather than its ℓ_2 norm. This is also what lets the bound saturate at the simplex diameter instead of growing without limit.

PROPOSITION 5.6 (THE BOUND IS TIGHT). *For every $\tau > 0$, $\theta > 0$, and $m \geq 2$, there exist a finite domain $\mathcal{T} \subset \mathbb{R}$, m candidates, and a threshold- θ policy such that*

$$S_2(H_t, P_\theta) = \sqrt{2} \tanh(\theta/2\tau)$$

for $m = 2$, and for any $\varepsilon > 0$,

$$S_2(H_t, P_\theta) \geq \sqrt{2} \tanh(\theta/2\tau) - \varepsilon$$

for $m > 2$.

PROOF SKETCH. Place two candidates at 0 and L and a protected pair

$$u = (L - \theta)/2, \quad v = (L + \theta)/2$$

symmetrically around their midpoint. Their distance is θ . With two candidates, the soft vote reduces to a logistic function; the two probability vectors differ in ℓ_2 norm by exactly $\sqrt{2} \tanh(\theta/2\tau)$. For $m > 2$, place the remaining candidates arbitrarily far away so that their softmax mass tends to zero. The two-candidate construction is then approached arbitrarily closely. The full calculation is in Appendix E. $\square$

No uniformly smaller bound depending only on reach and temperature is therefore possible. For small $r_d(G)/\tau$,

$$\sqrt{2} \tanh\left(\frac{r_d(G)}{2\tau}\right) \approx \frac{r_d(G)}{\sqrt{2}\tau},$$

while for large $r_d(G)/\tau$ it saturates at $\sqrt{2}$. The bound is worst-case over candidate populations; the exact sensitivity for a realized C_t can be substantially smaller. Section 4.4 explains why we nevertheless calibrate to the closed form.

Remark 1 (The soft bound stays below the hard vote's worst case). For $\tau > 0$ and finite reach, the bound is strictly below $\sqrt{2}$. This compares it with the hard vote's *worst case*, not with its exact policy-specific sensitivity: if no protected edge crosses a decision boundary, Hard has sensitivity 0. On an attribute with only one nonzero distance, such as a binary or flat categorical attribute, changing the graph cannot create finer distance levels; only τ moves the bound (Section 4.3).

5.5 Temperature and the Evolutionary Signal

Corollary 5.5 describes what a larger temperature buys in sensitivity, but not what it costs. ResampleMutate needs the released histogram to distinguish well-supported candidates from poorly supported ones. A nearly uniform vote carries little such signal even if it requires little noise.

PROPOSITION 5.7 (TEMPERATURE TRADES DISCRIMINATION FOR SENSITIVITY). Fix $C_t = \{c_1, \dots, c_m\}$.

(a) For x with margin

$$\gamma(x) = \min_{j \neq \text{nn}(x)} d(x, c_j) - d(x, c_{\text{nn}(x)}) > 0,$$

$$1 - v_{C_t}(x) [\text{nn}(x)] \leq (m-1)e^{-\gamma(x)/\tau}.$$

Thus the soft vote recovers the hard vote exponentially fast as $\tau \rightarrow 0^+$.

(b) With $\text{diam}(C_t) = \max_{j,k} d(c_j, c_k)$, for every x ,

$$\left\| v_{C_t}(x) - \frac{1}{m} \mathbb{1} \right\|_1 \leq 2 \tanh\left(\frac{\text{diam}(C_t)}{4\tau}\right) \leq \frac{\text{diam}(C_t)}{2\tau}.$$

Thus the vote approaches uniform at rate $O(1/\tau)$, governed by the candidate population rather than the policy.

PROOF SKETCH. For (a), divide the softmax numerator and denominator by the nearest candidate's term and bound every remaining exponential by $e^{-\gamma(x)/\tau}$. For (b), apply Lemma 5.4 to the uniform distribution with tilt $a_j = -d(x, c_j)/\tau$. The triangle inequality bounds the oscillation of these logits by $\text{diam}(C_t)/\tau$. $\square$

The two limits expose the trade-off. Small τ preserves a sharp nearest-neighbor signal but approaches the hard vote's sensitivity. Large τ reduces sensitivity but drives the vote toward uniform. At any fixed positive τ , the noiseless soft histogram also differs from the hard histogram, so the two methods can select toward different populations even when privacy noise becomes negligible. This is consistent with the empirical reversal at weaker privacy budgets in Section 6.

Remark 2 (No optimal temperature from the bounds). The sensitivity term in Corollary 5.5 depends only on the public policy, distance, and temperature. The utility cost of smoothing depends on the private data's margins, the candidate trajectory, and the privacy budget. The theoretical bounds therefore do not identify an optimal τ . Selecting it directly on private data would itself require privacy accounting, so we instead use the public-data pilot of Section 6.

Proposition 5.7 is deliberately not a convergence theorem. It characterizes the information retained by a single vote at a fixed candidate population. Whether the full population sequence converges under ResampleMutate depends also on mutation, resampling, and composition. The convergence analysis of the original hard-vote PE has itself required subsequent revision [8]. We therefore leave convergence under soft voting to future work; the sensitivity results above hold independently at every fixed round.

5.6 Floors on Sensitivity

Lemma 5.1 takes a maximum over all protected edges. An edge whose response does not shrink with the policy reach therefore imposes a floor on the achievable sensitivity reduction.

Remark 3 (Sensitivity floors). Two kinds of policy edges impose floors that no reach-based calibration can lower.

- (a) *Participation edges.* An edge $(u, \perp)$ is not controlled by the substitution distance d . With $\phi(\perp) = 0$, it contributes $\|\phi(u)\|$, so if participation edges are included for all values, then $S(f, P) \geq \max_u \|\phi(u)\|$. This equals 1 for the hard vote and is at most 1 for the soft vote, regardless of the substitution graph.
- (b) *Maximal-distance attributes.* Under a Gower-style distance, if attribute i contains a protected pair at its maximum distance $\text{diam}(\mathcal{T}_i)$, then Corollary 5.2 gives $r_d(G) \geq w_i \text{diam}(\mathcal{T}_i)$, and the analytical soft-vote bound is at least

$$\sqrt{2} \tanh\left(\frac{w_i \text{diam}(\mathcal{T}_i)}{2\tau}\right).$$

Complete graphs have this property. Under a flat categorical Gower term, so does any block with two or more distinct values, and a binary attribute admits no policy between empty and complete.

These floors do not imply that one policy is intrinsically worse than another for a fixed response function. They identify policy edges that the particular reach-dependent reduction of Corollary 5.5 cannot make smaller.

Scope of the analysis. The results in this section assume bounded replacement adjacency and unconstrained $\mathcal{I}_Q$. The main reduction is strongest when the protected substitutions have small reach under the same metric used by the vote, as with narrow numeric,

Table 2: Default per-attribute policy graphs.

Dataset	Attribute	Graph
Adult	age	threshold (7 years)
	hours.per.week	threshold (8 hours)
	education.num	threshold (2 levels)
	workclass	block (4 blocks)
	income	complete
Bank	age	threshold (7 years)
	balance	threshold (500)
	education	threshold (1 level)
	job	hierarchy (4 groups)
	y	complete

ordinal, or hierarchical policies. Participation edges and maximal-distance categorical edges can limit that reduction. The results are per-round sensitivity statements rather than convergence guarantees, but because they hold conditional on every realized candidate population, they compose through the adaptive evolutionary process described in Section 4.

6 Experiments

We ask four questions: when and why does soft voting outperform hard voting (Sections 6.2.1–6.2.3)? Can τ be chosen without private data (Section 6.2.4)? Which policies preserve the gain (Section 6.2.5)? Does the sensitivity bound hold, and what does calibration cost (Section 6.2.6)? Protocols and supporting results are in Appendix F.

6.1 Setup

Datasets. Adult [2] and Bank [22] each use five modeled columns including the label (Table 2), with 26,048 and 27,126 private training records. XOR draws k independent ordinal inputs $X_i \sim \text{Unif}\{0, \dots, 7\}$ and sets $Y = B_1 \oplus \dots \oplus B_k$, where $B_i = 1\{X_i \geq 4\}$, following Tab-PE’s high-order evaluation [24]. We use $k = 3$ (XOR-3) and $n = 12,000$.

Policies and reach. We use equal-weight Gower distance, $w = 1/p$, over the modeled columns ($p = 5$ on Adult and Bank, $p = 4$ on XOR-3). Numeric attributes are normalized by fixed domains: $[17, 90]$ for Adult age, $[18, 100]$ for Bank age, and $\{0, \dots, 7\}$ for X_i . A threshold θ then contributes reach $w\theta/\text{width}$, and multi-attribute policies take the maximum (Corollary 5.2). Except in Section 6.2.5, we protect one attribute: age with $\theta = 7$ on Adult and Bank (reach 0.0192 and 0.0171) and X_1 with one ordinal step on XOR-3 (reach $1/28$). Other modeled attributes lie outside the policy. Section 6.2.5 protects combinations up to all five columns; Appendix F.1 gives the rationale.

Methods. All methods use the same implementation of Algorithm 1, initial population, selection/mutation schedule, budget split, and paired randomness. Hard is the one-hot BF-Evo vote. In every checked round of the fixed-temperature runs, a protected edge crossed a nearest-candidate boundary, so its exact sensitivity was

$\sqrt{2}$, matching Tab-PE’s calibration; broader-policy runs use $\sqrt{2}$ directly. BF-Soft uses Corollary 5.5. We report its noise ratio

$$q = \sigma_S / \sigma_H = \tanh(r_d(G) / (2\tau)).$$

A standard-DP soft vote has full reach and, for the temperatures used in Section 6.2.3, effectively $q = 1$.

Measures and protocol. The primary error is mean ℓ_1 error over all two-way marginals (PWErr) on Adult and Bank, and ℓ_1 error over the 16-cell joint distribution of (B_1, B_2, B_3, Y) (HOErr) on XOR-3. Lower is better. Each release has 512 records over $T = 8$ rounds with equal budget allocation. We also report Blowfish ϵ at $\delta = 10^{-5}$ via Eq. (1); for example, $\rho = 10^{-4}, 6 \times 10^{-4}, 3 \times 10^{-3}$ give $\epsilon \approx 0.07, 0.17, 0.37$. Intervals are pointwise 95% Student- t intervals over paired runs. Best-of-sweep settings are descriptive and uncorrected for selection (Appendix F.2).

6.2 Results

6.2.1 Soft voting helps at strong privacy and reverses at weak privacy. Figure 2 compares Hard and BF-Soft at $\tau \in \{0.015, 0.03, 0.06\}$ over 17 budgets from 10^{-5} to 1, with twelve paired runs. Larger τ saves more noise: $q = 0.56, 0.31, 0.16$ on Adult; $0.51, 0.28, 0.14$ on Bank; and $0.83, 0.53, 0.29$ on XOR-3. In all nine comparisons, BF-Soft has lower mean error at $\rho = 10^{-5}$ and higher error at $\rho = 1$, with one sign change between them.

On Adult and Bank, interpolated crossings occur at $\rho \approx 5.5 \times 10^{-4} - 4.4 \times 10^{-3}$ ($\epsilon \approx 0.16 - 0.46$) and shift downward as τ grows (Appendix F.3). Thus more smoothing helps when privacy noise dominates but hurts once smoothing cost dominates, consistent with Section 5.5. XOR-3 crossings range more widely, from $\rho \approx 1.3 \times 10^{-4}$ to 0.36.

Training size has a similar effect. On XOR-3 at $\rho = 0.003$ and $\tau = 0.03$, mean error reduction is 22.8% at $n = 12,000$, 37.4% at 48,000, and –89.2% at 100,000 (Appendix F.9).

6.2.2 Temperature sweeps and the source of the gain. Figure 3 sweeps 20 temperatures from 0.00025 to 0.4 on Bank; Appendix F.5 gives Adult and XOR-3. In all nine dataset–budget panels, the lowest sampled BF-Soft mean occurs at an interior $\tau \in [0.002, 0.05]$ and is below Hard.

At $\rho = 10^{-4}$ ($\epsilon \approx 0.07$), the best tested settings reduce mean error from 0.696 to 0.427 on Adult and 0.976 to 0.730 on Bank (38.6% and 25.2%). The gain is not confined to one setting: Adult at 10^{-4} and Bank at 6×10^{-4} remain below Hard for every tested $\tau \in [0.004, 0.05]$. As ρ grows, the best τ falls: at $\rho = 3 \times 10^{-3}$ ($\epsilon \approx 0.37$), it is 0.006 on Adult and 0.002 on Bank, with 9.9% and 7.2% lower mean error. These best-of-sweep results are optimistic; Section 6.2.4 selects τ prospectively.

The corresponding noise ratios explain the source of the gain. At $\rho = 10^{-4}$, $q \approx 0.29$ on Adult and 0.17 on Bank, so most of Hard’s noise is removed. At 3×10^{-3} , $q \approx 0.92$ and 1.00: little or no noise is saved, and the remaining gain comes from the soft vote itself. A standard-DP soft vote would have essentially the same noise there. On XOR-3 the gain is smaller and less certain: 18.4% at 3×10^{-3} , while paired intervals include zero at 10^{-4} and 6×10^{-4} .

6.2.3 Noise savings versus smoothing cost. Let E_H and E_S be the errors of Hard and BF-Soft, and E_M the error of the soft vote with

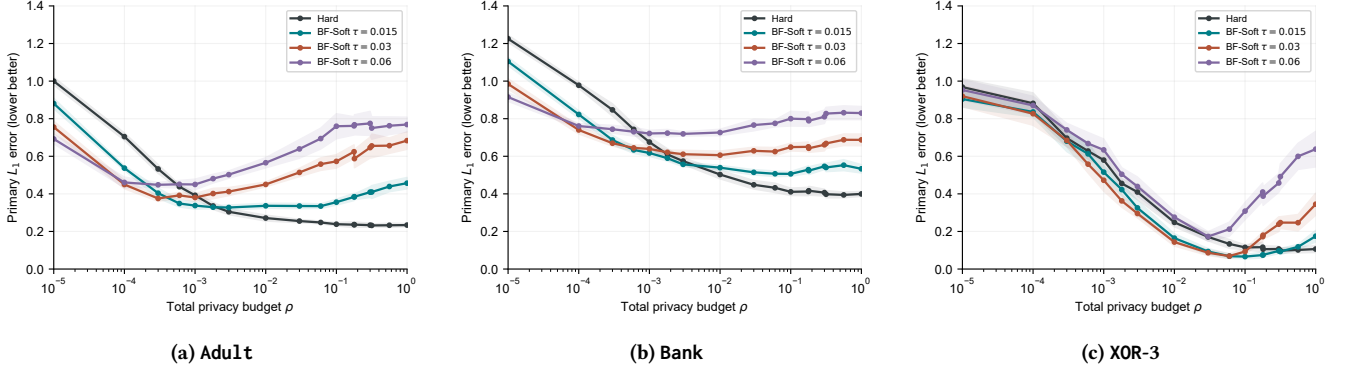


Figure 2: Error across privacy budgets at three fixed temperatures and default reach.

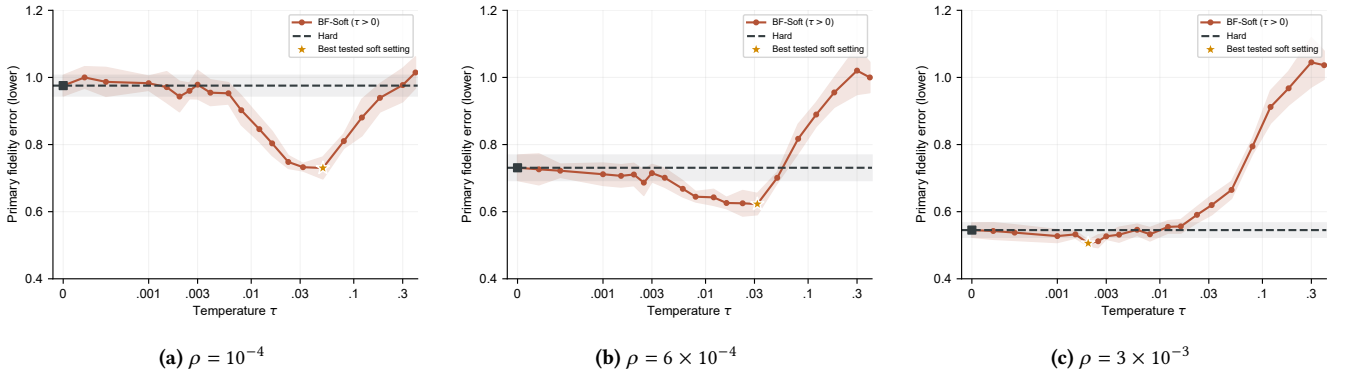
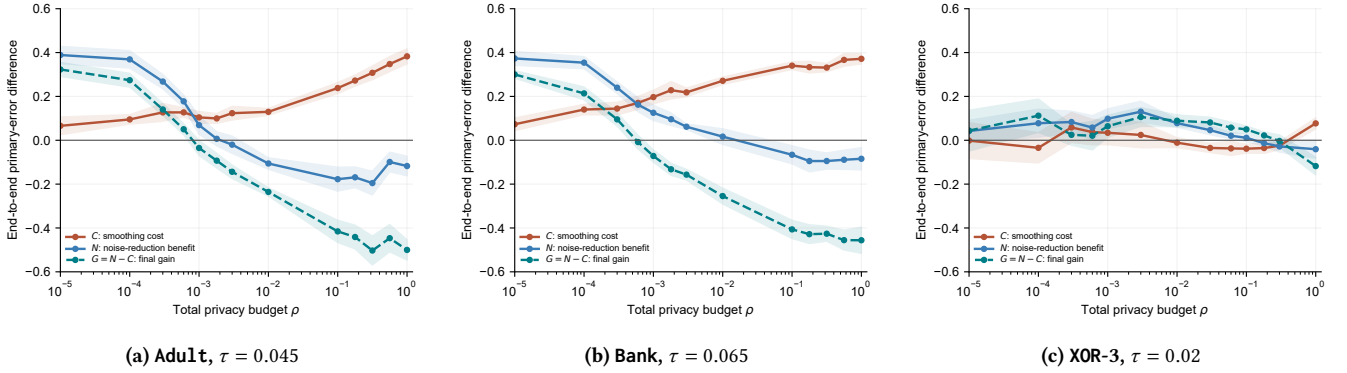


Figure 3: Temperature sweeps on Bank.

Figure 4: Smoothing cost C , noise-reduction benefit N , and final gain $G = N - C$ (Eq. (3)).

Hard's noise. Define

$$C = E_M - E_H, \quad N = E_M - E_S, \quad G = E_H - E_S = N - C. \quad (3)$$

Here C is the smoothing cost at matched noise, N the benefit of policy-specific noise reduction, and BF-Soft wins iff $N > C$. At these temperatures, E_M is also the error of the soft vote under standard DP.

Figure 4 reports these terms at one temperature per dataset over twelve paired runs. On Adult and Bank, $C > 0$ at every budget, so

standard-DP soft voting is worse than Hard; the strong-privacy gain comes entirely from policy-induced noise reduction. As ρ grows, N decreases while C grows, producing the crossover. On XOR-3 both effects are small; N first rises and then falls, consistent with severe noise and clipping limiting useful selection at the smallest budgets, although the decomposition does not isolate clipping.

6.2.4 Selecting τ without private data. The beneficial region depends jointly on τ , reach, and budget and is not determined by the

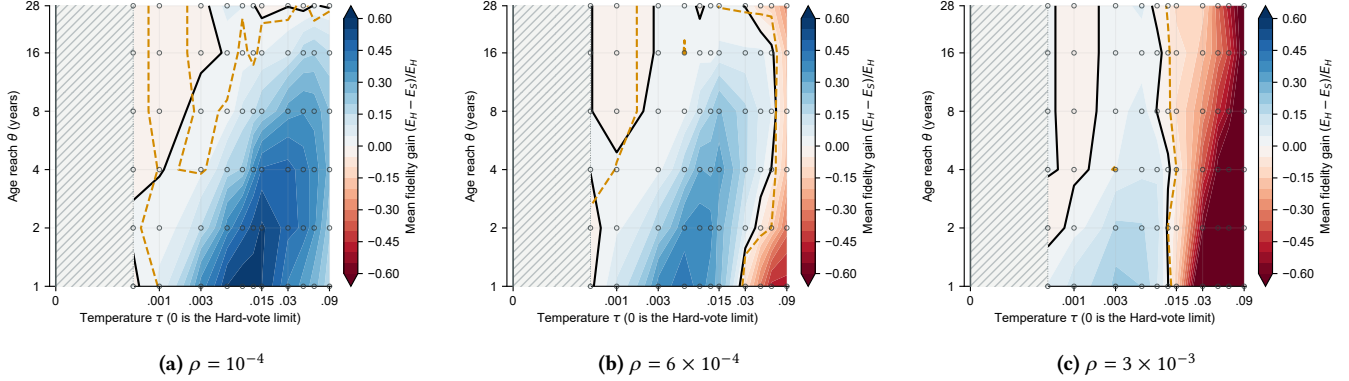


Figure 5: Relative gain on Adult over temperature and age reach. Black: observed zero-gain contour. Gold: public-pilot forecast, frozen before the target runs.

bound alone (Remark 2). We therefore test a public pilot. Each pilot uses 3,000 records disjoint from private training data; for Adult and Bank these are held-out validation records, while XOR-3 uses separately generated data. We rescale the pilot budget to match Gaussian noise in histogram proportions:

$$\rho_p = \rho_t \left(\frac{n_t}{n_p} \right)^2, \quad \frac{\sigma_p}{n_p} = \frac{\sigma_t}{n_t}, \quad (4)$$

for pilot and target sizes n_p, n_t . This matches normalized noise when sensitivity, output size, and round schedule agree, although finite-sample vote distributions and adaptive trajectories may differ.

On the original 210-setting Adult grid, the scaled pilot predicts the sign of mean gain in 190 cases (90.5%), versus 134 (63.8%) without scaling. Sign accuracy overstates performance near the zero-gain boundary and does not measure selected-setting utility. Figure 5 shows that the pilot contour tracks the target well at the two smaller budgets and less closely at 3×10^{-3} (Appendix F.7). Section 6.2.5 evaluates pilot-selected settings on Bank.

These pilots come from the same benchmark distribution as the target; performance under independently collected or shifted public data is untested. A public-only synthetic table is also a natural baseline when such data are available, but we did not evaluate it.

6.2.5 Which policies preserve the gain. Corollary 5.2 and Remark 3(b) predict which policies can benefit. Because reach is the maximum weighted per-attribute reach, adding narrow numeric attributes need not change it. A flat categorical or binary attribute raises it to that attribute’s full Gower weight, here 0.2, making $q \approx 1$ at useful temperatures. Figure 6 tests this prediction on both real datasets, with twelve paired runs per setting and Holm adjustment over 680 prespecified contrasts (Appendix F.8).

On Adult at $\tau = 0.03$, protecting age alone reduces mean paired error by 33.6% ([28.7, 38.4]%) at $\rho = 10^{-4}$, with $q = 0.31$. Adding hours.per.week leaves reach at 0.0192, so calibration, trajectories, and errors are unchanged. Numeric-only policies retain positive reductions; policies including workclass or income (reach 0.2) do not. Protecting all five columns gives $q = 0.9975$ and -12.0% ([−17.6, −6.5]%), not significant after Holm adjustment ($p_{\text{Holm}} = 0.174$). At larger budgets, this fixed-temperature BF-Soft is worse than Hard under every policy.

On Bank, a public pilot selects between Hard and nine temperatures before twelve target runs. Numeric-threshold policies retain a clear advantage at $\rho = 10^{-4}$: about 25% for age and age+balance, and 50% for balance alone, whose reach is much smaller. Adding education, job, or y reduces the gain below about 10%. With all five columns protected, the pilot selects $\tau = 0.003$ at $\rho = 0.003$, giving 5.8% reduction ([2.6, 9.1]%), but $q = 1$ shows that the gain comes from vote smoothing rather than lower noise and it does not survive Holm adjustment ($p_{\text{Holm}} = 0.613$). At $\rho = 10^{-4}$, $\tau = 0.001$ gives 1.4% ([−1.5, 4.3]%).

Thus q , computable from the policy before accessing private data, largely predicts whether noise reduction remains available. Narrow numeric policies with $q \ll 1$ preserve the strong-privacy gain; categorical or binary protection leaves little noise to save. Graded categorical distances may recover this benefit (Appendix D).

6.2.6 The calibration bound holds and is cheap. Figure 7a compares exact sensitivity with the analytical bound over 1,320 configurations of populations, geometries, seeds, and temperatures (Appendix F.2). Every measured BF-Soft sensitivity lies below the bound. Both the bound and median exact sensitivity fall with τ ; at $\tau \geq 20$ both are below 1% of $\sqrt{2}$. The bound is conservative on these populations.

Increasing reach has the opposite effect (Figure 7b): both analytical and exact sensitivity increase. Every plotted policy crosses a hard-vote boundary, so Hard remains at $\sqrt{2}$, whereas BF-Soft provides intermediate calibration levels.

Calibration is also cheap (Figure 7c). With $N = 8,192$ domain points and candidates, exact Hard and BF-Soft sensitivity take 211 ms and 1.48 s, respectively, versus about 2×10^{-4} ms for the bound. On product domains the gap is larger because exact calibration searches combinations of unprotected attribute values, while the bound’s cost is unchanged (Section 4.4).

7 Conclusion and Future Work

We asked whether grafting a Blowfish policy onto an evolutionary DP synthesizer improves its utility, and what determines when it does. The answer depends on the vote, not on the policy graph alone. For any linear query, policy-specific sensitivity is a maximum over the policy’s edges (Lemma 5.1). The hard vote’s sensitivity

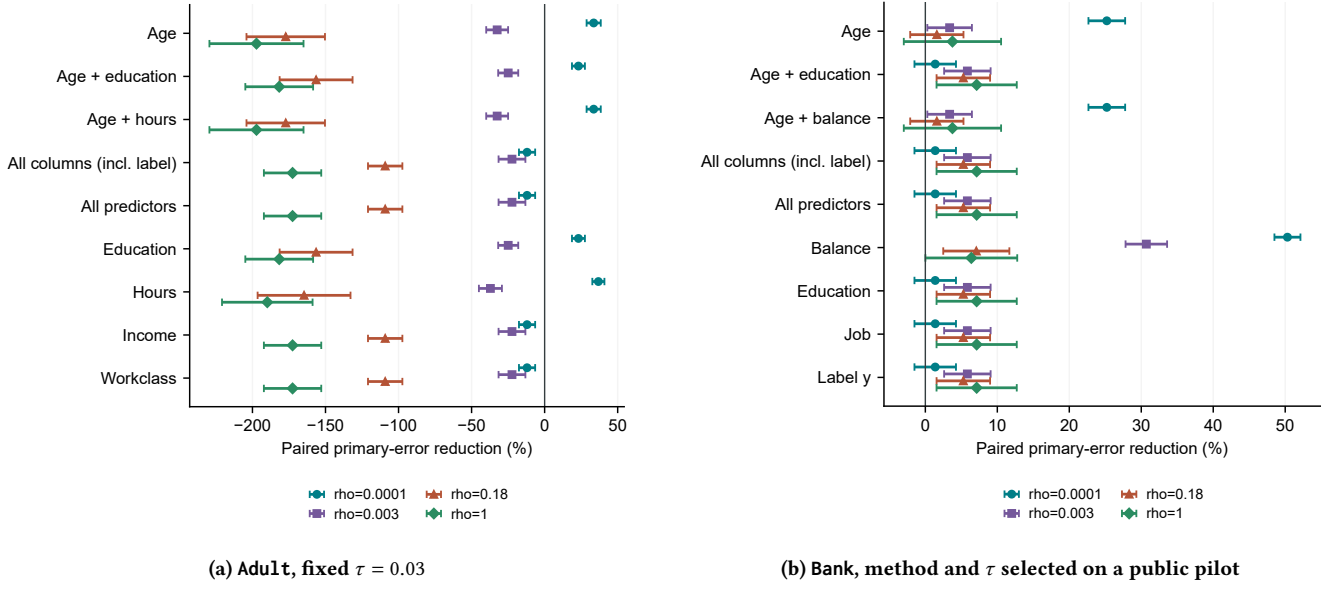


Figure 6: Paired error reduction relative to Hard (positive favors BF-Soft) under single- and multi-attribute policies.

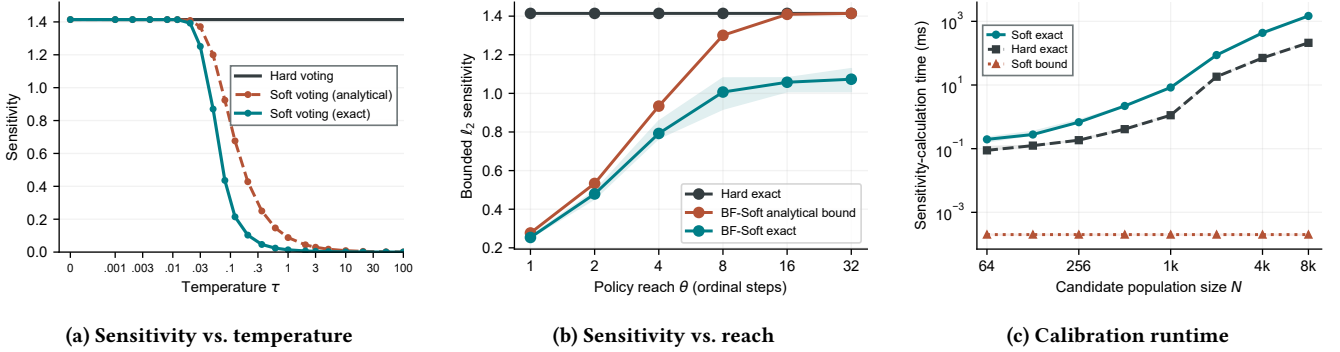


Figure 7: Exact sensitivity against the bound of Corollary 5.5, and calibration cost.

jumps to its worst case at the first protected boundary crossing, so a policy helps only when no protected edge crosses a boundary (Corollary 5.3). BF-Soft replaces it with a soft vote whose sensitivity has a closed-form bound in the policy reach and temperature, attained with two candidates and approached with more (Corollary 5.5, Proposition 5.6). Participation edges and maximal-distance protected attributes impose floors that this reach-dependent reduction cannot remove (Remark 3).

Temperature creates the corresponding utility trade-off: increasing it lowers sensitivity but weakens the vote’s ability to discriminate among candidates, and our bounds do not identify an optimal value (Remark 2). On real and synthetic data, we observe a beneficial region whose location shifts with policy reach and privacy budget. A public-data pilot predicts the sign of the gain on the original Adult grid with 90.5% accuracy without spending private budget (Section 6.2.4).

Our analysis covers linear queries under bounded, unconstrained adjacency. The experiments consider both single- and multi-attribute policies, with the strongest policy-induced noise

reductions under narrow numeric policies. We give no convergence analysis for BF-Soft under ResampleMutate, and the public pilot is validated only when pilot and target share a distribution.

A convergence analysis for BF-Soft and a regret guarantee for public-pilot temperature selection would address these limitations. The two relaxations of PE’s vote—WASP’s rank-based sharing and our distance-based smoothing—could also be combined. More broadly, the sensitivity identity applies to any linear query under a graph- or metric-based relaxation of DP [4, 14]. Whether other synthesis paradigms, such as marginal-based mechanisms or deep generative models trained under DP-SGD, admit a per-record response that tracks a policy’s notion of closeness remains open.

References

- [1] Miguel E. Andrés, Nicolás E. Bordenabe, Konstantinos Chatzikokolakis, and Catuscia Palamidessi. 2013. Geo-Indistinguishability: Differential Privacy for Location-Based Systems. In *ACM SIGSAC Conference on Computer and Communications Security (CCS)*. 901–914.
- [2] Barry Becker and Ronny Kohavi. 1996. Adult Data Set. UCI Machine Learning Repository. <https://archive.ics.uci.edu/dataset/2/adult>.
- [3] Mark Bun and Thomas Steinke. 2016. Concentrated Differential Privacy: Simplifications, Extensions, and Lower Bounds. In *Theory of Cryptography Conference (TCC)*. 635–658.
- [4] Konstantinos Chatzikokolakis, Miguel E. Andrés, Nicolás Emilio Bordenabe, and Catuscia Palamidessi. 2013. Broadening the Scope of Differential Privacy Using Metrics. In *Privacy Enhancing Technologies Symposium (PETS)*. 82–102.
- [5] Samuel N. Cohen and Eliana Fausti. 2023. Hyperbolic Contractivity and the Hilbert Metric on Probability Measures. *arXiv preprint arXiv:2309.02413* (2023).
- [6] Cynthia Dwork, Frank McSherry, Kobbi Nissim, and Adam Smith. 2006. Calibrating Noise to Sensitivity in Private Data Analysis. In *Theory of Cryptography Conference (TCC)*. 265–284.
- [7] Alessandro Epasto, Mohammad Mahdian, Vahab Mirrokni, and Manolis Zampetakis. 2020. Optimal Approximation–Smoothness Tradeoffs for Soft-Max Functions. In *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*. arXiv:2010.11450
- [8] Tomas González, Giulia Fanti, and Aaditya Ramdas. 2025. Private Evolution Converges. In *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv:2506.08312.
- [9] Samuel Haney, Ashwin Machanavajhala, and Bolin Ding. 2015. Design of Policy-Aware Differentially Private Algorithms. *arXiv preprint arXiv:1404.3722* (2015).
- [10] Xi He, Ashwin Machanavajhala, and Bolin Ding. 2014. Blowfish Privacy: Tuning Privacy-Utility Trade-offs Using Policies. In *Proceedings of the 2014 ACM SIGMOD International Conference on Management of Data (SIGMOD '14)*. ACM, 1447–1458.
- [11] Yingge Hu, Gautham Ramesh Babu, and Mostafa Milani. 2026. Soft Voting for Policy-Aware Private Data Synthesis (Extended Version). *arXiv preprint*, to appear.
- [12] Yingge Hu, Gautham Ramesh Babu, and Mostafa Milani. 2027. blowfish-evo-synth. <https://github.com/ayanamirei629/Soft-Voting-for-Policy-Aware-Private-Data-Synthesis>.
- [13] James Jordon, Jinsung Yoon, and Mihaela van der Schaar. 2019. PATE-GAN: Generating Synthetic Data with Differential Privacy Guarantees. In *International Conference on Learning Representations (ICLR)*.
- [14] Daniel Kifer and Ashwin Machanavajhala. 2014. Pufferfish: A Framework for Mathematically Defining Privacy. *ACM Transactions on Database Systems* 39, 1 (2014), 3:1–3:36.
- [15] Fragkiskos Koufogiannis, Shuo Han, and George J. Pappas. 2015. Optimality of the Laplace Mechanism in Differential Privacy. *arXiv preprint arXiv:1504.00065* (2015).
- [16] Zinan Lin, Sivakanth Gopi, Janardhan Kulkarni, Harsha Nori, and Sergey Yekhanin. 2024. Differentially Private Synthetic Data via Foundation Model APIs 1: Images. *International Conference on Learning Representations (ICLR)* (2024).
- [17] Karen Margaryan and Rachel Gupta. 2026. Ball Differential Privacy: How to Mitigate Data Reconstruction with Less Noise. *arXiv preprint arXiv:2607.04209* (2026).
- [18] Ryan McKenna, Gerome Miklau, and Daniel Sheldon. 2021. Winning the NIST Contest: A Scalable and General Approach to Differentially Private Synthetic Data. *Journal of Privacy and Confidentiality* 11, 3 (2021).
- [19] Ryan McKenna, Brett Mullins, Daniel Sheldon, and Gerome Miklau. 2022. AIM: An Adaptive and Iterative Mechanism for Differentially Private Synthetic Data. *Proceedings of the VLDB Endowment* 15, 11 (2022), 2599–2612.
- [20] Ryan McKenna, Daniel Sheldon, and Gerome Miklau. 2019. Graphical-Model Based Estimation and Inference for Differential Privacy. In *International Conference on Machine Learning (ICML)*. 4435–4444.
- [21] Frank McSherry and Kunal Talwar. 2007. Mechanism Design via Differential Privacy. In *48th Annual IEEE Symposium on Foundations of Computer Science (FOCS)*. IEEE, 94–103. doi:10.1109/FOCS.2007.66
- [22] Sérgio Moro, Paulo Cortez, and Paulo Rita. 2014. A Data-Driven Approach to Predict the Success of Bank Telemarketing. *Decision Support Systems* 62 (2014), 22–31. doi:10.1016/j.dss.2014.03.001
- [23] Vishnu Nair. 2026. Softmax is $1/2$ -Lipschitz: A Tight Bound Across All ℓ_p Norms. *Transactions on Machine Learning Research (TMLR)* (2026). arXiv:2510.23012.
- [24] Toan Tran, Arturs Backurs, Zinan Lin, Victor Reis, Li Xiong, and Sergey Yekhanin. 2026. Differentially Private Synthetic Data via APIs 4: Tabular Data. *arXiv preprint arXiv:2606.08259* (2026). To appear in ICML 2026.
- [25] Yidan Wang, Xin Chen, Wenjie Du, Bin Xiao, Cheng Zhang, and Ming Yan. 2026. Secret-Protected Evolution for Differentially Private Synthetic Text Generation. In *International Conference on Learning Representations (ICLR)*. arXiv:2510.10990.
- [26] Yonghui Xiao, Yilin Shen, Jinfei Liu, Li Xiong, Hongxia Jin, and Xiaofeng Xu. 2016. DPHMM: Customizable Data Release with Differential Privacy via Hidden Markov Model. *arXiv preprint arXiv:1609.09172* (2016).
- [27] Liyang Xie, Kaixiang Lin, Shu Wang, Fei Wang, and Jiayu Zhou. 2018. Differentially Private Generative Adversarial Network. *arXiv preprint arXiv:1802.06739* (2018).
- [28] Lei Xu, Maria Skoularidou, Alfredo Cuesta-Infante, and Kalyan Veeramachaneni. 2019. Modeling Tabular Data Using Conditional GAN. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- [29] Jun Zhang, Graham Cormode, Cecilia M. Procopiuc, Divesh Srivastava, and Xiaokui Xiao. 2017. PrivBayes: Private Data Release via Bayesian Networks. *ACM Transactions on Database Systems* 42, 4 (2017), 25:1–25:41.
- [30] Tianyuan Zou, Yang Liu, Peng Li, Yufei Xiong, Jianqing Zhang, Jingjing Liu, Xiaozhou Ye, Ye Ouyang, and Ya-Qin Zhang. 2025. WASP: Private Contrastive Evolution via Weighted, Multi-Candidate Voting for Synthetic Dataset Generation. *arXiv preprint arXiv:2502.00245* (2025).

A Extended Related-Work Differentiation

Section 3 gives the distinctions needed to position our contribution. This appendix expands three comparisons that are particularly close to the construction or analysis in this paper.

WASP’s rank-based weighting remains piecewise constant. WASP [30] replaces PE’s top-1 assignment with a multi-candidate vote whose weights depend on candidate rank. This changes how a record’s voting mass is distributed among candidates, but not how that mass responds continuously to the distance moved by a protected substitution.

For a fixed candidate population, each candidate’s rank is piecewise constant as a function of the record location. A substitution $x \rightarrow y$ that stays within the same ranking region leaves the rank-based weights unchanged; crossing a rank boundary changes one or more ranks discontinuously. This is the same structural phenomenon that makes the top-1 nearest-neighbor map piecewise constant over Voronoi cells (Section 5.3), although a top- Q rule can have more than the two response levels of the one-hot vote.

Thus rank weighting can smooth how mass is shared across candidates without making the response graded in the distance of the substitution. Our construction acts on this second axis: the softmax weights depend directly on candidate distances, so moving a record by a short distance induces a correspondingly bounded change in its vote. The two ideas are compatible; a mechanism could combine multi-candidate selection with a distance-sensitive weighting rule, which we leave to future work.

Ball-DP’s radius restriction targets a different object. Ball-DP [17] restricts standard DP’s neighbor relation to substitutions within a fixed radius of an underlying distance. At the level of adjacency, this corresponds to a threshold policy graph as in Definition 2.7. Its application, however, is convex empirical risk minimization rather than evolutionary synthesis.

Accordingly, the quantity that must be controlled is different. Ball-DP bounds how far the output of an optimization procedure can move after a radius-bounded perturbation of one training record. That change depends on the structure of the empirical-risk objective and its optimization. Our setting instead releases, at each round, an explicit aggregate of per-record votes. This makes the relevant quantity the norm of the change in a fixed record-level response $v_{C_i}(x)$, conditional on the current candidate population. Neither the voting construction, the candidate evolution, nor the closed-form soft-vote sensitivity bound studied here appears in Ball-DP.

The common point is the question posed by a radius-restricted neighbor relation: whether restricting the allowed substitution distance actually reduces the sensitivity of the object being released. The answer depends on that object’s response to the substitution. Our hard-vote result provides an example where the restriction can fail to give a graded reduction, while the soft vote is designed to exploit it.

Generic softmax bounds versus the policy-substitution bound. Temperature-scaled softmax is a well-studied smooth approximation to a hard selection rule. Generic Lipschitz bounds control

$$\|\text{softmax}(z) - \text{softmax}(z + a)\|$$

in terms of a norm of the arbitrary logit perturbation a ; related bounds arise both in work on the exponential mechanism [7, 21] and in direct analyses of the softmax map [23].

For our logits

$$z(x)_j = -d(x, c_j)/\tau,$$

the triangle inequality gives

$$|z(x)_j - z(y)_j| \leq d(x, y)/\tau$$

for every candidate j . A direct ℓ_2 treatment therefore gives

$$\|z(x) - z(y)\|_2 \leq \sqrt{m} d(x, y)/\tau,$$

which introduces a dependence on the candidate count m even though the perturbation has additional structure.

The important structure is that softmax is invariant to adding the same constant to all logits. Consequently, the absolute size of the logit displacement is not the relevant quantity; only its range,

$$\text{osc}(a) = \max_j a_j - \min_j a_j,$$

can change the normalized probabilities. For a policy substitution, Section 5.4 shows

$$\text{osc}(z(x) - z(y)) \leq 2d(x, y)/\tau,$$

with no dependence on m .

Lemma 5.4 then specializes the total-variation bound for exponential tilts of Cohen and Fausti [5] to obtain a nonlinear bound that saturates as the perturbation grows. After conversion from ℓ_1 to ℓ_2 , this gives Corollary 5.5,

$$S_2(H_t, P) \leq \sqrt{2} \tanh\left(\frac{r_d(G)}{2\tau}\right).$$

Unlike a linear Lipschitz upper bound, this expression never exceeds the $\sqrt{2}$ diameter relevant to two simplex vertices. Proposition 5.6 further shows that the bound is attained for two candidates and can be approached arbitrarily closely for any larger candidate population. Thus the contribution is not the use of softmax itself, but the policy-specific oscillation argument that yields a candidate-count-independent and tight calibration bound.

B PE’s Post-Hoc Threshold H

PE’s DP_NN_HISTOGRAM [16] takes a threshold H alongside the noise multiplier and distance function. After Gaussian noise is added to the vote histogram, entries of $\tilde{H}_t$ below H are zeroed before resampling. The purpose is practical: a hard-vote histogram over many candidates is sparse, and without thresholding, candidates whose counts are inflated primarily by noise may survive into

Table 3: Hard-vote sensitivity by adjacency notion when a change crosses a decision boundary.

Adjacency	effect of one change	ℓ_1	ℓ_2
add/remove (edge to $\perp$)	one vote appears or disappears	1	1
replacement, vote moves	vote moves from bin j to k	2	$\sqrt{2}$
replacement, same bin	vote does not move	0	0
k -row constrained neighbor	up to k aligned moves	$\leq 2k$	$\leq k\sqrt{2}$

ResampleMutate. H is tuned empirically in PE and is not adapted across rounds in [16].

H is post-processing. Every sensitivity result in Section 5 concerns the pre-noise histogram $H_t(D; C_t)$. Its policy sensitivity $S_2(H_t, P)$ determines the Gaussian noise scale required for a given privacy budget. Thresholding is applied only after the noisy histogram $\tilde{H}_t$ has been produced. It is therefore post-processing: it consumes no additional privacy budget and changes neither global nor policy-specific sensitivity. The same public threshold can thus be applied after Tab-PE, Hard, or BF-Soft without changing any privacy claim in Sections 4 and 5.

Retuning H under soft voting. Although thresholding does not affect privacy, its utility effect may depend on the vote. A hard-vote histogram is typically sparse, whereas a soft vote distributes each record’s unit mass across several candidates before noise is added. A threshold tuned for a sparse hard-vote histogram may therefore remove legitimate soft-vote mass, particularly at large τ . At small τ , the soft vote approaches the hard vote and the original thresholding behavior is recovered.

Whether H should be retuned specifically for soft voting is therefore an implementation and utility question rather than a privacy question. We do not analyze this choice here; all results in Section 5 concern $H_t(D; C_t)$ before either noise or post-processing.

C Algorithmic Comparison

Section 4.4 gives the main algorithmic comparison: Tab-PE, Hard/BF-Evo, and BF-Soft differ only in their per-record vote and sensitivity calibration. This appendix records two details that are useful but not needed for the main construction: the hard vote’s behavior under other adjacency notions and the cost of exact calibration on record-level policy graphs.

Other adjacency notions. Corollary 5.3 assumes bounded, unconstrained replacement adjacency, under which one record’s one-hot vote either stays in the same bin or moves from one bin to another. The corresponding ℓ_2 changes are 0 and $\sqrt{2}$.

With an add/remove edge $(u, \perp)$, one unit vote appears or disappears, giving both ℓ_1 and ℓ_2 change 1. More generally, public constraints can make two minimally different feasible databases differ in several rows. If at most k aligned record changes are required, triangle inequality gives the coarse bounds $2k$ in ℓ_1 and $k\sqrt{2}$ in ℓ_2 . Table 3 summarizes these cases. Our main analysis and experiments use bounded, unconstrained replacement adjacency.

Exact calibration for Hard. For a fixed candidate population C_t , Hard’s exact policy sensitivity requires determining whether any

protected edge $(u, v) \in E(G)$ satisfies

$$\text{nn}_{C_t}(u) \neq \text{nn}_{C_t}(v).$$

A single crossing certifies the value $\sqrt{2}$, so the search can terminate once one is found. To certify sensitivity 0, however, every protected edge must be checked. Because the population changes across rounds, this test must in principle be repeated for each C_t .

If nearest candidates are found by brute force, checking one edge costs $O(m)$ candidate-distance evaluations per endpoint. A complete scan therefore costs

$$O(|E(G)| \cdot \text{cost}(\text{nn}))$$

per round, where $\text{cost}(\text{nn}) = O(m)$ for a brute-force lookup and can be smaller when the metric space admits an appropriate index. The universal value $\sqrt{2}$ is always a valid constant-time fallback when exact policy calibration is not computed.

Record-level graph size. The relevant $E(G)$ above is the edge set of the *record-level* product graph, not merely the union of the per-attribute value graphs. Suppose an edge protects the pair (a, a') on attribute i . At record level, that pair induces an edge for every possible combination of the remaining attributes:

$$(x_1, \dots, a, \dots, x_p) \longleftrightarrow (x_1, \dots, a', \dots, x_p).$$

Consequently, even simple per-attribute policies can induce a record-level graph whose number of edges grows exponentially with the number of attributes. Exact BF-Soft sensitivity requires the same maximum over this graph, because Lemma 5.1 ranges over every protected record-level edge.

This distinction is why the analytical BF-Soft calibration is useful computationally. Once the weighted per-attribute reaches have been obtained, Corollary 5.2 gives the record-level reach by taking their maximum, and Corollary 5.5 evaluates the resulting sensitivity bound in constant time. No record-level graph needs to be enumerated.

Voting versus calibration cost. The voting step itself has the same asymptotic cost for all three methods. For each private record, both nearest-neighbor assignment and softmax voting require distances to the m current candidates, hence $O(m)$ work with a direct implementation. BF-Soft therefore does not add asymptotic voting cost relative to Tab-PE or Hard.

The difference lies in calibration. Tab-PE uses the fixed value $\sqrt{2}$. BF-Soft uses a bound computed once from $r_d(G)$ and τ . Hard’s exact policy calibration can instead require the round-dependent graph search described above. The experiment in Section 6.2.6 compares these calibration costs on a one-attribute domain; it does not measure end-to-end synthesis time.

D Distances for Categorical Attributes

Gower’s default categorical term treats every pair of distinct values alike: same value or not, with no nearer or farther mismatch. Under this term, a protected categorical attribute contributes its full Gower weight to the reach (Section 4.3, Remark 3). The remedy lies in d , not in the vote or the policy graph. If the attribute’s values carry a distance with more than one nonzero level, the soft vote and the bound of Corollary 5.5 apply unchanged, since neither assumes d is numeric, only that it is a metric.

When the values already fall into a taxonomy, such as job codes, product categories, or administrative regions, a natural choice is a *hierarchy*. This is a rooted tree over the values, with tree-path distance

$$d(u, v) = \text{depth}(u) + \text{depth}(v) - 2 \text{depth}(\text{lca}(u, v)),$$

the number of edges between u and v through their lowest common ancestor. This is a metric. With all leaves at depth L , it takes the nonzero values $2, 4, \dots, 2L$.

A threshold policy on this distance can therefore choose how deep a protected substitution may reach, from siblings under one parent up to the whole tree. After normalizing by the diameter $2L$, protecting only sibling substitutions gives the attribute reach w/L rather than w . For Bank’s job ($L = 2$), this halves the reach from 0.2 to 0.1. That is still large compared with a narrow numeric threshold such as age’s 0.017, so a hierarchy recovers gradation but not, at shallow depth, a small reach.

The flat block partition is the depth-1 case, with a single nonzero distance. A binary attribute admits no gradation under any distance, because a two-point domain has only one nonzero distance.

We use this construction only for Bank’s job, whose grouping is given in Appendix F.1. Other categorical attributes and the binary labels keep Gower’s default term, since they have no principled hierarchy to draw on.

E Full Proofs

This appendix gives the complete proofs of the results stated in Section 5. The main text includes the central derivations needed to understand the sensitivity bound; here we give the remaining algebra and equality conditions in full.

E.1 Proofs for Section 5.2

Lemma 5.1 (restated). For a linear query $f(D) = \sum_{x \in D} \phi(x)$, any policy graph $G = (\mathcal{T}, E)$, unconstrained I_Q , and bounded adjacency,

$$S_p(f, P) = \max_{(u, v) \in E} \|\phi(u) - \phi(v)\|_p,$$

with value 0 when $E = \emptyset$.

PROOF. Under bounded, unconstrained adjacency, a policy neighbor D' of D replaces exactly one record’s value u by some v such that $(u, v) \in E$, leaving every other record unchanged. Hence

$$f(D) - f(D') = \sum_{x \in D} \phi(x) - \sum_{x \in D'} \phi(x) = \phi(u) - \phi(v).$$

Conversely, every edge $(u, v) \in E$ can be realized by a pair of policy-neighboring databases containing the same remaining records. Taking the maximum over Definition 2.4 therefore gives

$$S_p(f, P) = \max_{(u, v) \in E} \|\phi(u) - \phi(v)\|_p.$$

If $E = \emptyset$, there are no protected substitution edges and the maximum is defined as 0. $\square$

Corollary 5.2 (restated). For a product domain $\mathcal{T} = \mathcal{T}_1 \times \dots \times \mathcal{T}_p$ with per-attribute graphs $G_i = (\mathcal{T}_i, E_i)$ and record-level product graph G ,

$$S_p(f, P) = \max_i S_{p,i}(f, P).$$

Under a Gower-style distance $d(x, y) = \sum_i w_i d_i(x_i, y_i)$,

$$r_d(G) = \max_i w_i r_{d_i}(G_i).$$

PROOF. By construction, every edge of G changes exactly one attribute. Write

$$E = \bigcup_{i=1}^p E'_i,$$

where E'_i contains the record-level edges that agree on every attribute except i and differ on attribute i by an edge of E_i . These sets are disjoint. By Lemma 5.1,

$$S_p(f, P) = \max_{(u,v) \in E} \|\phi(u) - \phi(v)\|_p = \max_i \max_{(u,v) \in E'_i} \|\phi(u) - \phi(v)\|_p,$$

which is the first claim.

For an edge $(u, v) \in E'_i$, all coordinates except i agree, so

$$d(u, v) = w_i d_i(u_i, v_i).$$

Maximizing first over E'_i and then over attributes gives

$$r_d(G) = \max_i w_i r_{d_i}(G_i).$$

□

Consequently, the analytical BF-Soft calibration depends only on the largest weighted per-attribute reach. Exact soft-vote sensitivity can still depend on the candidate population, and exact Hard sensitivity depends on whether current assignments cross a policy edge.

E.2 Proof for Section 5.3

Corollary 5.3 (restated). For the hard vote $\phi(u) = e_{\text{nn}(u)}$ and any policy graph $G = (\mathcal{T}, E)$,

$$S_2(f, P) = \sqrt{2}$$

if some $(u, v) \in E$ has $\text{nn}(u) \neq \text{nn}(v)$, and $S_2(f, P) = 0$ otherwise.

PROOF. If $\text{nn}(u) = \text{nn}(v)$, then

$$\phi(u) = \phi(v)$$

and the difference has norm 0.

Otherwise,

$$e_{\text{nn}(u)} - e_{\text{nn}(v)}$$

has exactly one coordinate equal to +1, one equal to -1, and all remaining coordinates equal to 0. Its ℓ_2 norm is therefore $\sqrt{2}$. Lemma 5.1 takes the maximum of this two-valued quantity over E . □

E.3 Proofs for Section 5.4

Lemma 5.4 (restated). Let p be a probability distribution on $\{1, \dots, m\}$ and

$$q_j = \frac{p_j e^{a_j}}{\sum_k p_k e^{a_k}}$$

with $\text{osc}(a) \leq s$. Then

$$\|p - q\|_1 \leq 2 \tanh(s/4),$$

and the bound is attained by a suitable two-point distribution.

This is a specialization of the Hilbert-projective-metric bound of Cohen and Fausti [5]. For completeness we give the elementary argument.

PROOF. Adding the same constant to all a_j leaves q unchanged. We may therefore center them so that

$$a_j \in [-s/2, s/2].$$

Let

$$W_j = e^{a_j}, \quad L = e^{-s/2}, \quad U = e^{s/2},$$

so $W_j \in [L, U]$ and $LU = 1$. Define

$$Z = \sum_j p_j W_j = \mathbb{E}_p[W].$$

Then

$$q_j - p_j = p_j \frac{W_j - Z}{Z}.$$

Because

$$\sum_j p_j (W_j - Z) = 0,$$

the positive and negative parts have equal total mass, hence

$$\|p - q\|_1 = \frac{2}{Z} \mathbb{E}_p[(W - Z)^+].$$

For any $w \in [L, U]$, write

$$w = \alpha(w)U + (1 - \alpha(w))L, \quad \alpha(w) = \frac{w - L}{U - L}.$$

The function

$$g_Z(w) = (w - Z)^+$$

is convex. Since $L \leq Z \leq U$, its values at the endpoints are 0 and $U - Z$. Convexity therefore gives the chord bound

$$(w - Z)^+ \leq \alpha(w)(U - Z).$$

Taking expectation,

$$\mathbb{E}_p[(W - Z)^+] \leq \frac{Z - L}{U - L} (U - Z).$$

Thus

$$\|p - q\|_1 \leq \frac{2}{Z} \frac{(Z - L)(U - Z)}{U - L}.$$

Using $LU = 1$, this becomes

$$\|p - q\|_1 \leq \frac{2}{U - L} \left[(U + L) - \left(Z + \frac{1}{Z} \right) \right].$$

By AM-GM,

$$Z + \frac{1}{Z} \geq 2,$$

with equality at $Z = 1$. Since $1 \in [L, U]$, this value is feasible and maximizes the upper bound. Therefore

$$\|p - q\|_1 \leq \frac{2(U + L - 2)}{U - L}.$$

Substituting $U = e^{s/2}$ and $L = e^{-s/2}$,

$$\frac{2(U + L - 2)}{U - L} = \frac{2(\cosh(s/2) - 1)}{\sinh(s/2)} = 2 \tanh(s/4).$$

For equality, the chord bound requires W to be supported only on $\{L, U\}$, and the second step requires $Z = 1$. Choosing probabilities so that

$$\Pr(W = L) = \frac{U - 1}{U - L}, \quad \Pr(W = U) = \frac{1 - L}{U - L}$$

satisfies both conditions. Hence a two-point distribution attains the bound. □

Corollary 5.5 (restated). For the soft-vote histogram $H_t(D; C_t)$ and any policy graph G ,

$$S_2(H_t, P) \leq \sqrt{2} \tanh\left(\frac{r_d(G)}{2\tau}\right),$$

independent of $m = |C_t|$. For a threshold policy G_θ , this becomes

$$S_2(H_t, P_\theta) \leq \sqrt{2} \tanh(\theta/2\tau).$$

PROOF. By Lemma 5.1, it suffices to consider one policy edge $(x, y) \in E$. Let

$$z(x)_j = -d(x, c_j)/\tau, \quad z(y)_j = -d(y, c_j)/\tau.$$

For each candidate,

$$|z(x)_j - z(y)_j| = \frac{|d(x, c_j) - d(y, c_j)|}{\tau} \leq \frac{d(x, y)}{\tau}$$

by the reverse triangle inequality. Hence

$$\text{osc}(z(y) - z(x)) \leq \frac{2d(x, y)}{\tau}.$$

Moreover,

$$v_{C_t}(y)_j = \frac{v_{C_t}(x)_j \exp(z(y)_j - z(x)_j)}{\sum_k v_{C_t}(x)_k \exp(z(y)_k - z(x)_k)},$$

so $v_{C_t}(y)$ is precisely an exponential tilt of $v_{C_t}(x)$. Applying Lemma 5.4 with

$$s = \frac{2d(x, y)}{\tau}$$

gives

$$\|v_{C_t}(x) - v_{C_t}(y)\|_1 \leq 2 \tanh\left(\frac{d(x, y)}{2\tau}\right).$$

Let

$$z = v_{C_t}(x) - v_{C_t}(y).$$

Because both votes sum to 1, $\sum_j z_j = 0$. Writing $z = z^+ - z^-$, the positive and negative parts have equal ℓ_1 mass, say a , so

$$\|z\|_1 = 2a.$$

Also,

$$\|z\|_2^2 = \|z^+\|_2^2 + \|z^-\|_2^2 \leq \|z^+\|_1^2 + \|z^-\|_1^2 = 2a^2 = \frac{\|z\|_1^2}{2}.$$

Therefore

$$\|z\|_2 \leq \frac{\|z\|_1}{\sqrt{2}}.$$

Combining the two bounds,

$$\|v_{C_t}(x) - v_{C_t}(y)\|_2 \leq \sqrt{2} \tanh\left(\frac{d(x, y)}{2\tau}\right).$$

Finally, $d(x, y) \leq r_d(G)$ for every policy edge, so maximizing over E proves the claim. $\square$

Proposition 5.6 (restated). For every $\tau > 0$, $\theta > 0$, and $m \geq 2$, there exist a finite domain $\mathcal{T} \subset \mathbb{R}$, m candidates, and a threshold- θ policy such that

$$S_2(H_t, P_\theta) = \sqrt{2} \tanh(\theta/2\tau)$$

for $m = 2$, and for every $\varepsilon > 0$,

$$S_2(H_t, P_\theta) \geq \sqrt{2} \tanh(\theta/2\tau) - \varepsilon$$

for $m > 2$.

PROOF. Fix $L \geq \theta$. Let

$$c_1 = 0, \quad c_2 = L,$$

and define

$$u = \frac{L - \theta}{2}, \quad v = \frac{L + \theta}{2}.$$

Then

$$d(u, v) = \theta,$$

so (u, v) is a protected edge under a threshold- θ policy. Also,

$$d(u, c_2) - d(u, c_1) = \theta, \quad d(v, c_2) - d(v, c_1) = -\theta.$$

For $m = 2$, the first coordinate of the soft vote is

$$v_{C_t}(x)_1 = \frac{e^{-d(x, c_1)/\tau}}{e^{-d(x, c_1)/\tau} + e^{-d(x, c_2)/\tau}} = \sigma\left(\frac{d(x, c_2) - d(x, c_1)}{\tau}\right),$$

where σ is the logistic function. Hence

$$v_{C_t}(u)_1 = \sigma(\theta/\tau), \quad v_{C_t}(v)_1 = \sigma(-\theta/\tau).$$

The second coordinates are complementary, so

$$\|v_{C_t}(u) - v_{C_t}(v)\|_2 = \sqrt{2} [\sigma(\theta/\tau) - \sigma(-\theta/\tau)].$$

Using

$$\sigma(z) - \sigma(-z) = \tanh(z/2),$$

we obtain

$$\|v_{C_t}(u) - v_{C_t}(v)\|_2 = \sqrt{2} \tanh(\theta/2\tau).$$

Corollary 5.5 provides the matching upper bound, so equality holds.

For $m > 2$, place the remaining candidates sufficiently far from u and v . For example, take

$$c_k = L + R + k, \quad k = 3, \dots, m.$$

As $R \rightarrow \infty$,

$$e^{-d(x, c_k)/\tau} \rightarrow 0$$

uniformly for $x \in \{u, v\}$ and $k \geq 3$. The first two coordinates of both votes therefore converge to the two-candidate softmax, while the remaining coordinates converge to 0. Consequently,

$$\|v_{C_t}(u) - v_{C_t}(v)\|_2 \rightarrow \sqrt{2} \tanh(\theta/2\tau).$$

For any prescribed $\varepsilon > 0$, some finite R achieves a value within ε , proving the claim. $\square$

E.4 Proof of Proposition 5.7

Proposition 5.7 (restated). Fix $C_t = \{c_1, \dots, c_m\}$.

(a) For x with margin

$$\gamma(x) = \min_{j \neq \text{nn}(x)} d(x, c_j) - d(x, c_{\text{nn}(x)}) > 0,$$

$$1 - v_{C_t}(x)[\text{nn}(x)] \leq (m - 1)e^{-\gamma(x)/\tau}.$$

(b) With

$$\text{diam}(C_t) = \max_{j, k} d(c_j, c_k),$$

for every x ,

$$\left\|v_{C_t}(x) - \frac{1}{m} \mathbf{1}\right\|_1 \leq 2 \tanh\left(\frac{\text{diam}(C_t)}{4\tau}\right) \leq \frac{\text{diam}(C_t)}{2\tau}.$$

PROOF. For part (a), let $c^* = c_{nn(x)}$. Then

$$v_{C_t}(x)[nn(x)] = \frac{1}{1 + \sum_{j \neq nn(x)} \exp\left(-\frac{d(x, c_j) - d(x, c^*)}{\tau}\right)}.$$

By definition of $\gamma(x)$,

$$d(x, c_j) - d(x, c^*) \geq \gamma(x)$$

for every $j \neq nn(x)$. Therefore

$$v_{C_t}(x)[nn(x)] \geq \frac{1}{1 + (m-1)e^{-\gamma(x)/\tau}}.$$

Hence

$$1 - v_{C_t}(x)[nn(x)] \leq \frac{(m-1)e^{-\gamma(x)/\tau}}{1 + (m-1)e^{-\gamma(x)/\tau}} \leq (m-1)e^{-\gamma(x)/\tau}.$$

For part (b), let

$$p = \frac{1}{m} \mathbb{1}, \quad a_j = -d(x, c_j)/\tau.$$

Tilting p by a gives exactly $q = v_{C_t}(x)$. Let c_a and c_b be respectively the farthest and nearest candidates from x . Then

$$d(x, c_a) - d(x, c_b) \leq d(c_a, c_b) \leq \text{diam}(C_t)$$

by the triangle inequality. Thus

$$\text{osc}(a) \leq \frac{\text{diam}(C_t)}{\tau}.$$

Lemma 5.4 gives

$$\left\| v_{C_t}(x) - \frac{1}{m} \mathbb{1} \right\|_1 \leq 2 \tanh\left(\frac{\text{diam}(C_t)}{4\tau}\right).$$

Finally, $\tanh z \leq z$ for $z \geq 0$, yielding

$$\left\| v_{C_t}(x) - \frac{1}{m} \mathbb{1} \right\|_1 \leq \frac{\text{diam}(C_t)}{2\tau}.$$

□

Scope of Proposition 5.7. The proposition concerns one vote at one fixed candidate population. It does not characterize the evolution of $C_1, C_2, \dots$ under mutation and resampling, nor does it imply convergence of the overall synthesizer. Those questions depend on the full ResampleMutate dynamics. The sensitivity results of Section 5, in contrast, hold conditional on each fixed population regardless of whether the population sequence converges. The convergence analysis for the original hard-vote PE has itself required subsequent refinement [8]; we therefore leave a corresponding analysis for soft voting to future work.

F Additional Experimental Details and Results

This appendix gives the policy rationale, experimental protocol, and supporting results for Section 6.

F.1 Policy Rationale

We follow the style of Blowfish’s motivating examples [10]: protect a narrow range of substitutions while allowing coarser distinctions to remain exposed.

For the numeric and ordinal attributes:

- The age and hours.per.week thresholds blur a small number of years or hours without collapsing the full scale.

- education.num on Adult and education on Bank protect nearby schooling levels.
- balance’s threshold of 500 protects balances that are relatively close while leaving coarse differences, such as overdrawn versus comfortably positive, distinguishable.

The binary labels income and y admit no nontrivial intermediate graph: once their two distinct values are protected, the attribute graph is complete. Protecting either therefore contributes the attribute’s full Gower weight to the record-level reach (Remark 3).

workclass: flat blocks. workclass on Adult has nine values. We use a flat block partition into private, government, self-employed, and other, following the type of categorical grouping used in Blowfish [10]. We do not add another hierarchy level. For example, assigning a different policy distance to Federal-gov and Local-gov would require an additional privacy rationale not supplied by the attribute itself. Thus workclass illustrates a categorical policy with coarse blocks but no graded within-block distance.

job: hierarchical distance. job on Bank has twelve values and supports a more structured grouping. We use a two-level hierarchy with four broad labor-market groups below the root:

- professional/technical: *management, technician, admin;*
- manual/service: *blue-collar, services, housemaid;*
- independent: *self-employed, entrepreneur;*
- not in the labor force: *retired, student, unemployed, unknown.*

The twelve original categories form the leaves.

The job field is not coded using an external occupational taxonomy, such as SOC, so this grouping is our construction rather than an externally defined hierarchy. This is consistent with the Blowfish model, in which the data owner specifies which distinctions are protected.

We use the tree-path distance of Appendix D. Two leaves in the same parent group have distance 2 (leaf–parent–leaf), while leaves in different groups have distance 4 (leaf–parent–root–parent–leaf). We normalize by the tree diameter 4 before using this distance in Gower distance.

A threshold policy on this tree has two nontrivial levels: $\theta = 2$ protects substitutions within the same broad group, while $\theta = 4$ protects every pair of job categories. This gives job a genuine graded distance that the flat workclass representation lacks.

XOR-3. The XOR-3 experiments protect X_1 only. Each input takes eight ordinal values, $0, \dots, 7$. The default policy protects one-step substitutions, while the temperature–reach experiments vary the protected width from one to four steps. One step is $1/7$ of the input’s span, approximately 14.3%. With four equally weighted modeled columns, it contributes record-level reach $1/28$. The XOR-3 settings are therefore not defined by a common percentage threshold and are not matched to the normalized reach or graph density of Adult or Bank.

F.2 Protocol and Reporting

Implementation. All methods use the same implementation of Algorithm 1, equal-weight normalized Gower distance, and hybrid selection. The standard experimental configuration uses 512 candidates in rounds 1–3 and 1,024 candidates in rounds 4–8. Each run

releases 512 records over eight rounds, with the privacy budget divided equally across rounds. Random streams are paired across methods.

Repetitions.

- Fixed-temperature budget sweeps use twelve paired runs at 17 budgets from 10^{-5} to 1.
- Temperature sweeps use six seeds at 20 temperatures from 0.00025 to 0.4.
- The smoothing-cost/noise-benefit decomposition uses twelve runs.
- Matched-noise experiments use six seeds at 25 budgets.
- XOR-3 temperature-reach experiments use six paired runs per setting, cycling through three generated populations.
- Public-pilot experiments use three pilot seeds and six paired target seeds.
- Broader-policy experiments use twelve paired runs.
- The XOR-3 training-size experiment uses twelve independent data/algorithm seed pairs, nested training prefixes, and a common 10,000-record test reference.

Intervals and tests. Intervals are pointwise 95% Student- t intervals, not simultaneous intervals. On *Adult* and *Bank* they quantify algorithmic randomness for a fixed data split. For paired run s , relative error reduction is

$$R_s = 100 \frac{E_{H,s} - E_{S,s}}{E_{H,s}},$$

and we report its mean $\bar{R}$. Thus we average the run-level relative reductions rather than taking the ratio of the grand means. Positive values favor BF-Soft.

Tests of zero mean paired reduction in Section 6.2.5 are two-sided. We apply Holm correction over 680 prespecified primary contrasts. Results chosen retrospectively as the best point in a parameter sweep are descriptive and are not corrected for parameter selection.

Sensitivity measurements.

- Figure 7a uses a domain of eight binary attributes with single-bit substitution edges. It covers four population sizes (32–256), three candidate geometries, five seeds, and 22 temperatures including $\tau = 0$, for 1,320 measurements. Each plotted curve is the median over the 60 configurations at a fixed temperature.
- Figure 7b uses a 64-point ordinal domain, 20 candidates, six policy reaches, and twelve candidate seeds. Bands show interquartile ranges.
- Figure 7c varies domain size and candidate count together over $N \in \{64, 128, \dots, 8192\}$ with five paired seeds. Exact computations evaluate the same $N \times N$ distance matrix and scan the policy edges. These timings measure calibration only, not voting, selection, mutation, or the full evolutionary loop.

F.3 Fixed-Temperature Crossings

Table 4 reports where the sampled mean errors of BF-Soft and Hard change order in Figure 2. Each value interpolates the mean difference linearly in $\log \rho$ between two neighboring measured

budgets. No value extrapolates outside the experimental grid. Consequently, these are descriptive between-sample crossing estimates, not directly observed crossing budgets.

Table 4: Noise ratio q , interpolated crossing budget $\rho_\times$, and corresponding ϵ at $\delta = 10^{-5}$.

Dataset	τ	q	$\rho_\times$	ϵ
Adult	0.015	0.56	2.0×10^{-3}	0.31
	0.03	0.31	1.1×10^{-3}	0.23
	0.06	0.16	5.5×10^{-4}	0.16
Bank	0.015	0.51	4.4×10^{-3}	0.46
	0.03	0.28	1.6×10^{-3}	0.27
	0.06	0.14	6.7×10^{-4}	0.18
XOR-3	0.015	0.83	0.36	4.5
	0.03	0.53	0.12	2.5
	0.06	0.29	1.3×10^{-4}	0.08

F.4 Matched Noise Ratios

Figure 8 asks whether the noise ratio q alone determines fidelity. Within each dataset, we hold q fixed while varying policy reach and temperature together.

For *Adult* we use

$$(\tau, \theta) \in \{(0.015, 2), (0.03, 4), (0.06, 8)\}.$$

For *Bank* we set

$$\tau = \frac{r_d(G)}{2 \operatorname{atanh}(q)}$$

at $\theta \in \{2, 4, 8\}$, giving

$$(\tau, \theta) \in \{(0.01335, 2), (0.02671, 4), (0.05341, 8)\}.$$

Both datasets have $q = 0.1806$.

For XOR-3 we use

$$(\tau, \theta) \in \{(0.01, 1), (0.02, 2), (0.04, 4)\},$$

with reaches 1/28, 2/28, and 4/28, respectively. All three have $q = 0.9453$. Thus q is matched within the XOR-3 panel but not across datasets. Each panel uses 25 budgets from 10^{-5} to 1 and six paired seeds.

The soft-vote curves differ despite equal noise ratios. Thus q alone does not determine fidelity: temperature also changes the vote and the subsequent evolutionary trajectory. Because the matched settings protect different substitution ranges, this experiment isolates equal-noise behavior rather than comparing mechanisms under identical privacy policies.

F.5 Temperature Sweeps on *Adult* and XOR-3

Figure 9 gives the *Adult* and XOR-3 temperature sweeps corresponding to the *Bank* panels in Figure 3. The numerical summaries in Section 6.2.2 use the lowest sampled BF-Soft mean in each panel.

For XOR-3 at $\rho = 3 \times 10^{-3}$, the best tested temperature is $\tau = 0.023$, with $q \approx 0.65$. At $\rho = 10^{-4}$, the absolute improvement at the best tested setting is 0.0022, illustrating that a positive relative gain can still correspond to a small absolute difference on this workload.

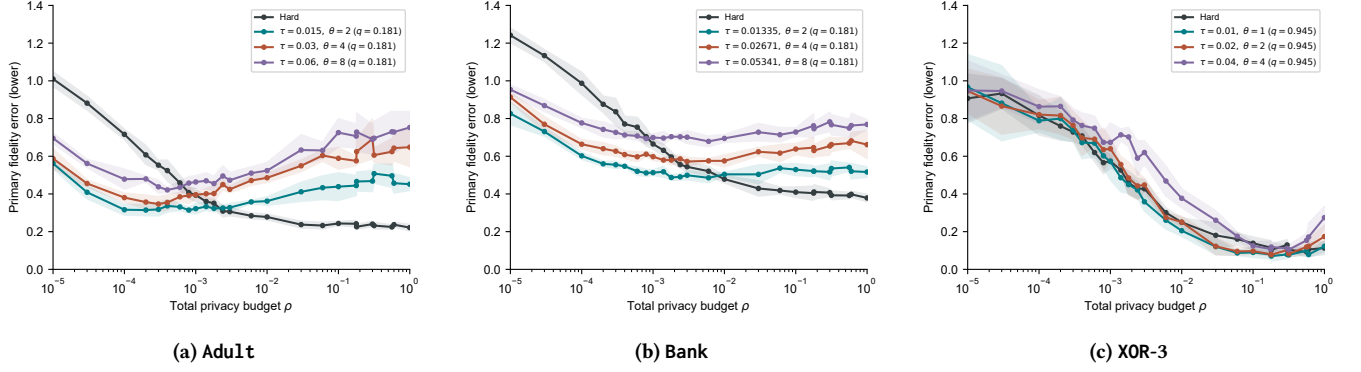


Figure 8: Error across privacy budgets at matched noise ratios.

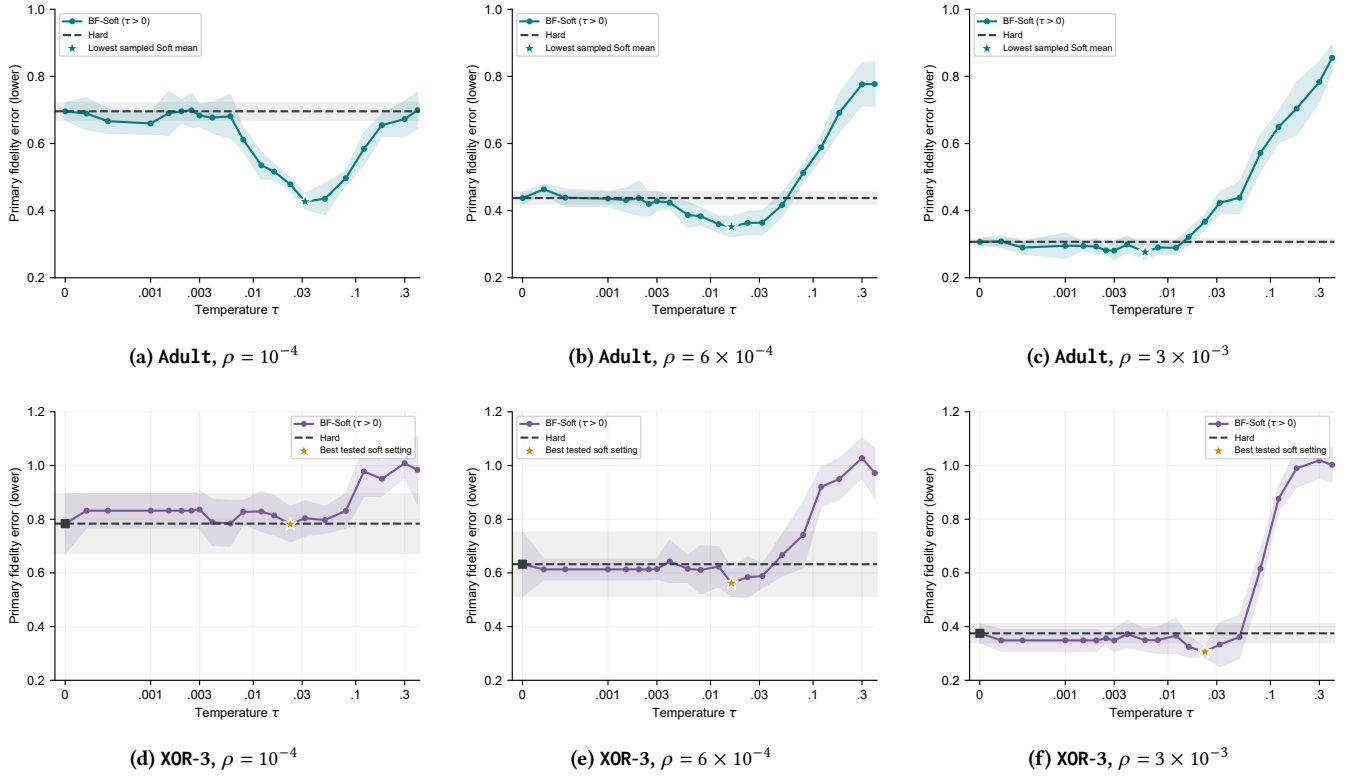


Figure 9: Temperature sweeps on Adult and XOR-3.

F.6 Temperature–Reach Interaction on XOR-3

Figure 10 jointly varies τ and the protected width $\theta \in \{1, 2, 3, 4\}$ at three privacy budgets, with 48 evaluated settings per panel.

The largest mean relative gains are 39.8%, 40.4%, and 19.9% at $\rho = 0.03, 0.18$, and 1 , respectively ($\epsilon \approx 1.2, 3.1$, and 7.8):

- at $\rho = 0.03$, the largest measured mean gain occurs at $\tau = 0.003$ for the tied reaches $\theta \in \{2, 3, 4\}$;
- at $\rho = 0.18$ and $\rho = 1$, it occurs at $(\tau, \theta) = (0.01, 3)$.

At each of these settings,

$$\frac{r_d(G)}{2\tau} > 5,$$

so $q \approx 1$. The policy therefore saves essentially no noise. The observed improvement instead comes from the effect of soft voting on selection for this high-order workload. A soft vote calibrated to standard DP would receive essentially the same noise at these settings.

Only integer reaches from one to four steps were evaluated, and we did not separately sample temperatures between the hard-vote limit $\tau = 0$ and the smallest positive tested temperature.

F.7 Public-Pilot Forecasts

The public-data pilot in Section 6.2.4 uses 3,000 records disjoint from the private training sample. For `Adult` and `Bank` these are held-out records from the same benchmark distribution; `XOR-3` uses independently generated public data.

The sign-accuracy result reported in Section 6.2.4 uses the original 210-setting `Adult` grid. A later extension added lower temperatures to the plotted grid in Figure 5; those additional cells are not included in the reported 190/210 sign-accuracy calculation.

The forecast agrees most closely with the observed target boundary at the two smaller privacy budgets. A contour is absent when the mean gain does not change sign on the evaluated grid. Where the forecast and target contours disagree, the pilot predicts the wrong direction of the mean difference.

The pilots come from the same benchmark distribution as the target. These results therefore test whether a held-out public sample can guide temperature selection under matched data generation, not whether the procedure is robust to independently collected or shifted public data.

F.8 Broader Policies

The broader-policy experiments vary which of the selected attributes are protected while keeping the modeled schema, Gower weights, and evaluation workload fixed. A record-level policy edge still changes only one coordinate. Protecting several attributes therefore means taking the union of their single-attribute substitution edges; it does not permit arbitrary simultaneous changes to multiple attributes.

These experiments concern the five modeled columns in Table 2, not every column in the original UCI datasets.

`Adult`. At $\tau = 0.03$ and $\rho = 10^{-4}$, the `age` and `hours.per.week` policies use thresholds of seven years and eight hours. Their weighted reaches are 0.0192 and 0.0163, respectively. Protecting both therefore leaves the record-level reach at 0.0192, the same as protecting age alone. Consequently, the analytical calibration and paired trajectories are unchanged.

Adding attributes with larger reaches eventually changes this behavior. Protecting all five modeled columns raises the reach to 0.2, increasing q from 0.3092 to 0.9975 and leaving almost no policy-specific noise reduction.

`Bank`. For the broader-policy `Bank` experiment, temperature is selected using a public pilot before target evaluation. The pilot contains 3,000 validation records and uses a disjoint reference drawn from the remaining validation records. Its budget is scaled according to Eq. (4).

For each policy, the selector chooses among:

- `Hard`;
- $\tau \in \{0.001, 0.003, 0.015, 0.03, 0.06, 0.12, 0.3, 0.6\}$;
- the temperature giving $q = 0.2$ for that policy.

Selection minimizes mean public error. Ties favor `Hard` and then the smaller temperature. No target errors are used during selection.

As discussed in Section 6.2.5, numeric-threshold policies preserve substantial strong-privacy gains, while adding attributes with larger categorical or binary reach reduces the available noise saving. Pilot-selected gains under the broadest policies do not survive the Holm correction and should be interpreted as exploratory.

F.9 Private Training Size

Figure 11 varies the `XOR-3` private training size over

$$n \in \{3,000, 12,000, 48,000, 100,000\}.$$

The experiment keeps three inputs, 512 released records, $\tau = 0.03$, and one-step protection of X_1 , giving

$$r_d(G) = 1/28, \quad q = 0.5337.$$

At $\rho = 0.003$, the mean paired error reduction is:

- 22.83% ([15.45, 30.22]%) at $n = 12,000$;
- 37.37% ([25.19, 49.55]%) at $n = 48,000$;
- -89.18% ([-122.06, -56.30]%) at $n = 100,000$.

Between $n = 12,000$ and $n = 100,000$, `Hard`'s mean error falls from 0.389 to 0.118, while `BF-Soft`'s falls from 0.297 to 0.214. Both methods therefore improve with more private data, but `Hard` improves faster.

The privacy budget is fixed across training sizes. Because the histogram noise scale does not grow proportionally with n , its size relative to the empirical histogram shrinks as n increases. Larger n therefore acts similarly to moving toward a weaker-noise regime, where the cost of smoothing can dominate. This experiment concerns one-step protection of X_1 at fixed parity order and does not imply the same trend for every policy or workload.

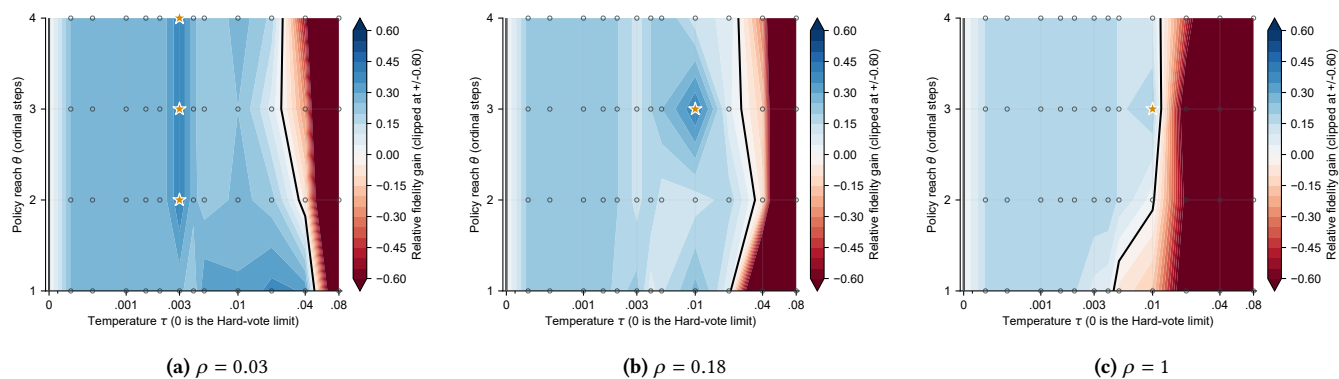


Figure 10: XOR-3 relative gain over temperature and policy reach. Black contours interpolate zero mean gain.

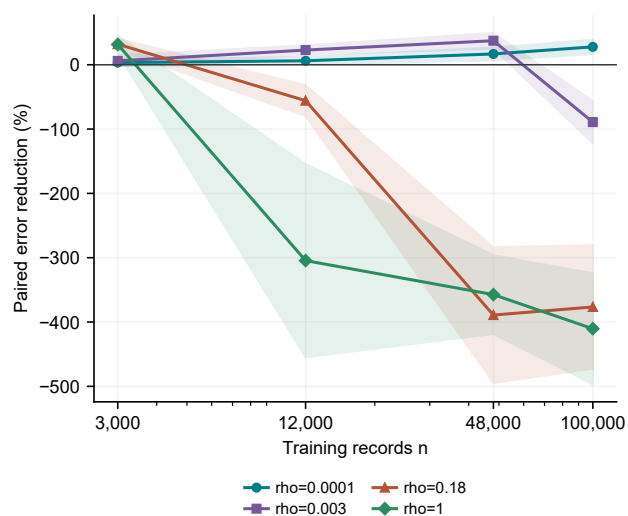


Figure 11: XOR-3 paired error reduction relative to Hard across private training sizes.